Knowledge-Based Systems 299 (2024) 111998

Contents lists available at ScienceDirect

Knowledge-Based Systems

journal homepage: www.elsevier.com/locate/knosys

Constrained multi-objective optimization problems: Methodologies, algorithms and applications

Yuanyuan Hao [a,d], Chunliang Zhao [b], Yiqin Zhang [b], Yuanze Cao [b,*], Zhong Li [c,*]

[a] Faculty of Mathematics and Computer Science, Fernuniversität in Hagen, Hagen, 58097, Germany
[b] School of Data Science, Qingdao University of Science and Technology, No. 99, Songling Road, Qingdao, 266061, China
[c] School of Mathematics and Statistics, Minnan Normal University, China
[d] School of Systems Science, Beijing Jiaotong University, No. 3, Shangyuan Village, Beijing, 100028, China

ARTICLE INFO

Keywords:
Constrained multi-objective optimization problems
Evolutionary algorithms
Machine learning
Applications

ABSTRACT

Constrained multi-objective optimization problems (CMOPs) are widespread in practical applications such as engineering design, resource allocation, and scheduling optimization. It is high challenging for CMOPs to balance the convergence and diversity due to conflicting objectives and complex constraints. Researchers have developed a variety of constrained multi-objective optimization algorithms (CMOAs) to find a set of optimal solutions, including evolutionary algorithms and machine learning-based methods. These algorithms exhibit distinct advantages in solving different categories of CMOPs. Recently, constrained multi-objective evolutionary algorithms (CMOEAs) have emerged as a popular approach, with several literature reviews available. However, there is a lack of comprehensive-view survey on the methods of CMOAs, limiting researchers to track the cutting-edge investigations in this research direction. Therefore, this paper reviews the latest algorithms for handling CMOPs. A new classification method is proposed to divide literature, containing classical mathematical methods, evolutionary algorithms and machine learning methods. Subsequently, it reviews the modeling and algorithms of CMOPs in the context of practical applications. Lastly, the paper gives potential research directions with respect to CMOPs. This paper is able to provide guidance and inspiration for scholars studying CMOPs.

1. Introduction

Constrained multi-objective optimization problems are widely present in scientific research and practical engineering, including optimal wireless sensor location design [1], wind farm layout optimization [2], and portfolio optimization [3]. Without loss of generality, CMOPs are described as follows [4],

$$\begin{aligned} \text{minimize} \quad & \vec{F}(\vec{x}) = [f_1(\vec{x}), f_2(\vec{x}), \dots, f_k(\vec{x})]^T \\ \text{subject to} \quad & g_i(\vec{x}) \le 0, \quad \text{for } i = 1, 2, \dots, M \\ & h_j(\vec{x}) = 0, \quad \text{for } j = 1, 2, \dots, L \\ & \vec{x} = [x_1, x_2, \dots, x_D]^T \in \mathrm{S}, \end{aligned} \tag{1}$$

where S is the decision space and $\vec{x}$ is the D-dimension decision vector, $\vec{x} \in \mathrm{S}$, $\vec{F}(\vec{x})$ is the objective vector that contains k objectives, $f_i(\vec{x})$ is the i-th objective function, $g_i(\vec{x})$ and $h_j(\vec{x})$ are the inequality and equality constraint functions, M and L are the numbers of inequality constraint functions and equality constraint functions.

The total constraint violation degree of the decision variables quantifies the extent to which the constraints are violated by the solution. This degree can be denoted as [5–7],

$$CV(\vec{x}) = \sum_{i=1}^{M} \max\{0, g_i(\vec{x})\} + \sum_{j=1}^{L} \max\{0, |h_j(\vec{x})| - \sigma\}, \tag{2}$$

where σ is a positive tolerance value to relax the equality constraints. If $CV(\vec{x}) = 0$, then $\vec{x}$ is a feasible solution (FS); otherwise, $\vec{x}$ is an infeasible solution (IFS). All the feasible solutions form the feasible region.

For any two solutions $\vec{x}_1$ and $\vec{x}_2$, if $f_i(\vec{x}_1) \le f_i(\vec{x}_2)$ for each $i \in \{1, \dots, k\}$ and $f_j(\vec{x}_1) < f_j(\vec{x}_2)$ for at least one $j \in \{1, \dots, k\}$, then $\vec{x}_1$ dominates $\vec{x}_2$; otherwise, $\vec{x}_1$ and $\vec{x}_2$ are non-dominated solutions. A set of non-dominated feasible solutions is referred to as the Pareto optimal set (POS). Each solution within the POS is a Pareto optimal solution. The projection of POS in the objective space forms the Pareto front (PF). The PF of a constrained multi-objective optimization problem is typically denoted as CPF, while that of an unconstrained problem is

* Corresponding authors.
E-mail addresses: 19114003@bjtu.edu.cn (Y. Hao), zhaochunliang@qust.edu.cn (C. Zhao), 4023112051@mails.qust.edu.cn (Y. Zhang), yuanzecao@mails.qust.edu.cn (Y. Cao), zhong.li@fernuni-hagen.de (Z. Li).

https://doi.org/10.1016/j.knosys.2024.111998
Received 3 April 2024; Received in revised form 9 May 2024; Accepted 22 May 2024
Available online 29 May 2024

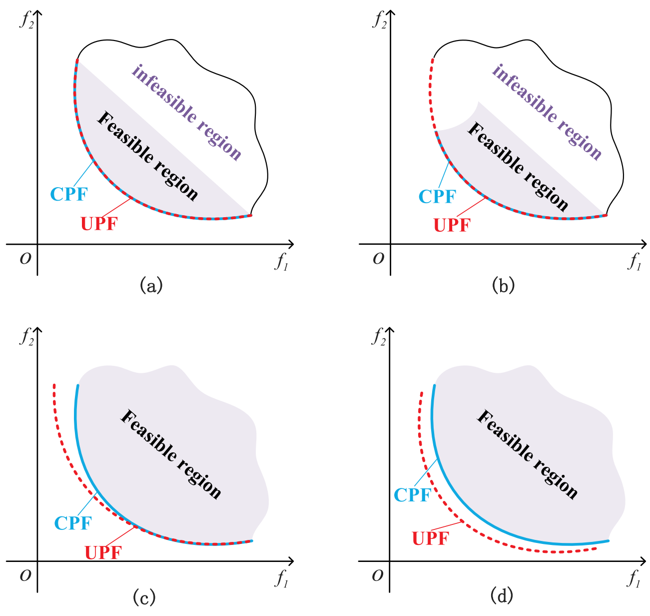

Fig. 1. The illustrations show four types of CMOPs.

denoted as UPF. Existing CMOPs can be broadly categorized into four types based on the relationship between CPF and UPF [8]:

- Type I. CPF and UPF overlap completely, and UPF is fully feasible, as shown in Fig. 1(a);
- Type II. CPF is a subset of UPF, and UPF is partially feasible, as shown in Fig. 1(b);
- Type III. CPF partially overlaps with UPF, and UPF is partially feasible, as shown in Fig. 1(c);
- Type IV. CPF is completely separate from UPF, and UPF is completely unfeasible, as shown in Fig. 1(d).

The task of solving CMOPs is to identify a set of Pareto non-dominated solutions that exhibit convergence and uniform distribution. However, the difficulty of achieving this objective is significant [9]:

1. Feasibility: The constraints may render a large portion of the search space infeasible, thereby complicating the task of locating a small feasible region, as shown in Fig. 2(a).
2. Diversity: The discontinuity of the feasible region results in the true CPF exhibiting a fractured characteristic. This fracture in the CPF complicates the algorithm's ability to cover its entire range, making it a challenge to obtain a complete CPF, as shown in Fig. 2(b).
3. Convergence: The solution is trapped in the external feasible region, making it difficult to towards the true CPF across the infeasible region, as shown in Fig. 2(c).

Facing the aforementioned challenges, it is crucial to achieve a balance between convergence, diversity, and feasibility when solving CMOPs. In the past three decades, methods based on classical mathematical optimization were first introduced to constrained multi-objective optimization [11,12]. However, constrained multi-objective evolutionary algorithms have gradually replaced them due to their greater efficiency in dealing with nonlinear or non-convex problems [13].

CMOEAs are heuristic-based methods that combine constraint handling techniques (CHTs) and multi-objective evolutionary algorithms. CHTs are used to deal with constraints, while multi-objective evolutionary algorithms are the main driving algorithm used to approximate the PF. However, it is difficult for methods based on evolution to better extract the characteristics of the problem, and thus cannot adaptively adjust strategies or CHTs to flexibly solve complex and changing CMOPs via the characteristics of the problem. Methods based on machine learning have become a new trend expected to overcome the drawbacks of evolution-based methods with the emergence of the new wave of machine learning.

The current review on CMOPs focuses on CMOEAs, and the methods involved are not comprehensive [14]; therefore, this paper reviews mathematical methods, evolutionary algorithms, and machine learning perspectives to form a new systematic review. The contribution of this paper can be summarized as follows:

- This paper more scientifically and comprehensively categorizes advanced CMOAs into three categories: (1) methods based on classical mathematics [15,16]; (2) methods based on evolution [17,18]; (3) methods based on machine learning [19,20]; as shown in Fig. 3. Each category is introduced and analyzed in detail, and the advantages and disadvantages of various CMOAs are also summarized. Some promising future research directions will be discussed.
- This paper introduce the application of constrained multi-objective optimization in some typical fields more comprehensively than previous studies. These cases provide guidance to relevant practitioners in selecting suitable optimization algorithms.
- While the current CMOAs have demonstrated strong performance in solving many conventional CMOPs, there are still challenges arising from the increasing complexity of CMOPs. These challenges include how to better extract features from CMOPs, how to autonomously recommend CMOAs, how to handle dynamic interval CMOPs, and how to handle large-scale CMOPs.

The rest of this paper is organized as follows: Section 2 classifies the existing CMOAs and discusses the advantages and disadvantages of each type of algorithm, Section 3 presents some real-world applications of CMOAs, Section 4 discusses the future trends and challenges for constrained multi-objective optimization.

2. State-of-the-art CMOAs

This section provides a comprehensive review of the current research landscape of CMOAs, delineating their respective strengths and weaknesses. Based on their fundamental design principles and solution strategies, CMOAs are categorized into three distinct categories. Collectively, these categories encompass the current state-of-the-art CMOAs.

2.1. Methods based on classical mathematics

These methods are typically based on classical mathematical optimization theories, including linear programming [21], ε-constraint methods [22], and fuzzy optimization theory [23]. These algorithms apply classical mathematical optimization theories in innovative ways to tackle complex real-world problems. By combining different mathematical theories, these algorithms can provide more robust and efficient solutions.

Mavrotas [15] introduced an augmented ε-constraint method (AUGMECON), which precludes the generation of weak Pareto optimal solutions and expedites the process. This method eliminates superfluous iterations and efficaciously applies the ε-constraint method to yield Pareto optimal solutions in multi-objective mathematical programming (MOMP). Liu et al. [24] combined the selection-based constraint handling with fuzzy membership functions, and constructed a new constraint handling method–multi-objective fuzzy selection, which is used to select two candidate schemes from a given group, and the selection rules are as follows: (1) Given two feasible solutions, select the one with better objective function value; (2) Given two infeasible solutions,

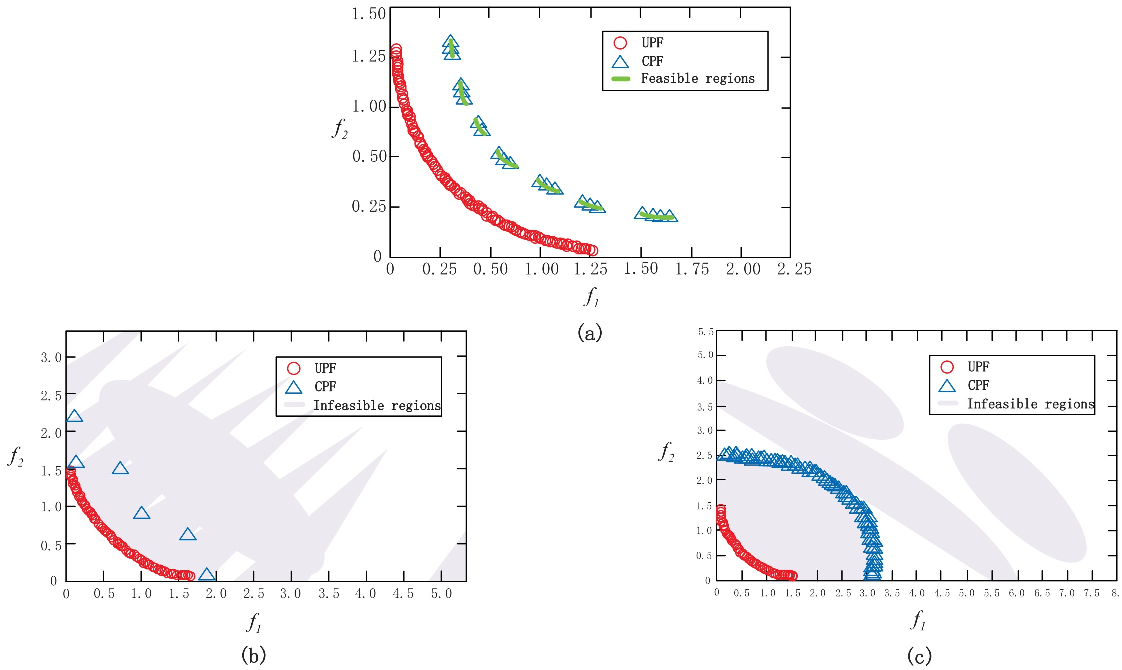

Fig. 2. Illustrations for three different types of difficulties [10]: (a) LIR-CMOP4, (b) LIR-CMOP11, (c) LIR-CMOP7.

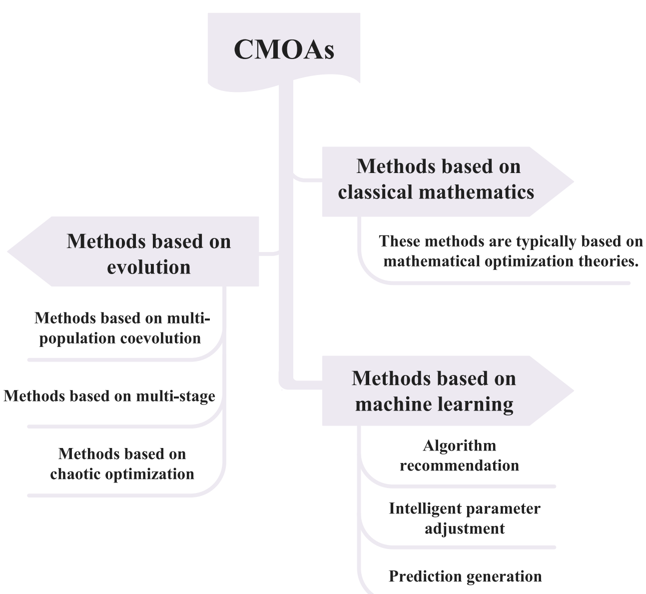

Fig. 3. State-of-the-art CMOAs.

select the one with smaller violation of constraint conditions; (3) If one solution is feasible and the other is not, select the FS. Saha et al. [25] proposed a novel fuzzy rule-based penalty function approach for single-objective nonlinear constrained optimization problems, a fuzzy inference system based on Mamdani-type IF-THEN rules, which does not require complex explicit mathematical formulas, and designed a parameter-free robust penalty function for constraint handling. Li et al. [26] proposed a fuzzy search bias (FSB) guided CHT, where the search bias is guided by a membership function, which is used to satisfy fuzzy knowledge, that is, solutions that violate larger constraints tend to search for solutions that violate smaller constraints, and solutions that violate smaller constraints tend to search for solutions with better objective function values.

Moreover, Yang et al. [16] devised an algorithm for fuzzy multi-objective linear fractional programming (FMOLFP) issues, employing the superiority and inferiority measure method (SIMM). This method handles fuzzy numbers within constraints to maximize the degree of membership and introduces linear objective programming for problems with fuzzy fractional objectives. Cocchi and Lapucci [27] proposed an enhancement to the classical generalized Lagrangian method, targeting equality and inequality constrained multi-objective optimization problems to improve computational efficiency. Ansary and Panda [28] developed a globally convergent sequential quadratic programming (SQP) approach for multi-objective optimization with inequality constraints, securing a feasible descent direction through linear approximation of all objectives and constraints, and addressing constraint violations with a non-differentiable penalty function. Yang et al. [29] transformed a nonlinear optimization model into a mixed-integer linear programming problem via linearization, applied the augmented generalized ε-constraint method for solution, and employed fuzzy clustering to select the final solution from the Pareto optimal set based on decision-maker preferences. Han et al. [30] proposed a new fuzzy constraint handling technique based on [24], and introduced a new concept called "fuzzy dominance", which accurately describes and quantifies the comprehensive difference between solutions from the perspective of objectives and constraints, and embedded the proposed fuzzy constraint handling technique into a decomposition-based multi-objective evolutionary algorithm as an implementation.

Besides, Lai et al. [31] established multi-objective approximate gradient projection (MAGP) and linear multi-objective approximate gradient projection (LMAGP) sequential optimality conditions without constraint qualifications for multi-objective constrained optimization problems. Pérez-Cañedo et al. [32] extended the concepts of dominance and Pareto optimality to the intuitionistic fuzzy domain, using a dictionary ordering criterion for triangular intuitionistic fuzzy numbers (TIFNs) and an intuitionistic fuzzy ε-constraint approach for multi-objective linear programming with TIFNs. Sharma et al. [33] proposed a method for a multi-objective two-layer chance-constrained optimization problem in an intuitionistic fuzzy environment, representing technological coefficients and objective function coefficients with Triangular Intuitionistic fuzzy numbers and resource coefficients with normal random variables, utilizing component-wise optimization to convert intuitionistic fuzzy objectives into their deterministic equivalents. Alouane and Boulif [34] proposed a fuzzy inference engine-based method that can switch between different forms of transformation functions (TFs) for selecting the best constraint ordering. TFs are a type of constraint-handling approach that operates in the phenotypic space. The TF adaptively adjusts its definition by incorporating the constraint order according to the priority of the constraints. The fuzzy inference engine then selects the most suitable form of the TF based on the problem characteristics and the constraint complexity, thus improving the solution efficiency and quality.

In general, methods based on classical mathematics rely on established optimization theories and require exact mathematical models and deterministic algorithms to solve multi-objective optimization problems. Although precise and appropriate for problems with certain properties like differentiability and convexity, they may struggle with complex, nonlinear, or non-convex challenges.

2.2. Methods based on evolution

These methods explore the solution space to identify feasible solutions by mimicking natural evolutionary processes. They are divided into three categories.

2.2.1. Methods based on multi-population coevolution

These methods typically rely on two or three populations, each assigned a different task. The populations communicate with each other to achieve a better balance of convergence, feasibility, and diversity. One population acts as the main population and performs a constrained search to find feasible solutions and the CPF, while the other populations act as auxiliaries and perform an unconstrained search to find the UPF. Additionally, auxiliary populations can aid the main population in escaping from infeasible regions or local optima by transferring knowledge. This method can be classified as weak or strong coevolution, depending on the level of coevolution among the populations.

(1)Strong coevolution: We use a two-population example, in which the two populations are typically merged before the mating selection stage and undergo their own mating selection processes. The populations are then combined with their own offspring and those of other populations, sharing all the offspring generated by all populations, as illustrated in Fig. 4(a). This method enhances the cooperation ability of the population, utilizes the information between the populations fully, but may also result in population homogenization and premature convergence.

Ming et al. [35] introduced a dual-population approach to handle infeasible solutions; the main population applies an adaptive penalty, while the secondary employs constrained non-dominated sorting genetic algorithm II(C-NSGA-II) to modify the objective vector of such solutions. During mating selection, both populations combine and utilize binary tournament selection for choosing parents. This algorithm efficiently leverages IFS data, yet overlooks Pareto optimal solutions at the constraint edge. Liu et al. [36] proposed a bidirectional coevolution (BiCo) algorithm to explore the feasible region boundary from both sides, helping to find Pareto optimal solutions at the boundary. Restrictive mating selection based on CV and angle density promotes strong coevolution. Jiao et al. [37] proposed a multiform optimization (MFO) algorithm that addresses the complexity of CMOPs by focusing on Pareto solutions at constraint boundaries and search space partitioning. The algorithm divides the problem into dual tasks: the main and auxiliary populations perform constraint and ϵ-constraint searches, with the ϵ boundary decreasing over iterations. Based on [37], Qiao et al. [38] proposed a double-balanced evolutionary multi-task optimization (DBEMTO) algorithm, which uses an adaptive mechanism to achieve the balance between intra-task and inter-task, enabling the population to utilize suitable knowledge, and also uses an effective environmental selection method to exploit the information of parents and offspring, enhancing the effect of strong coevolution. Li et al. [39]

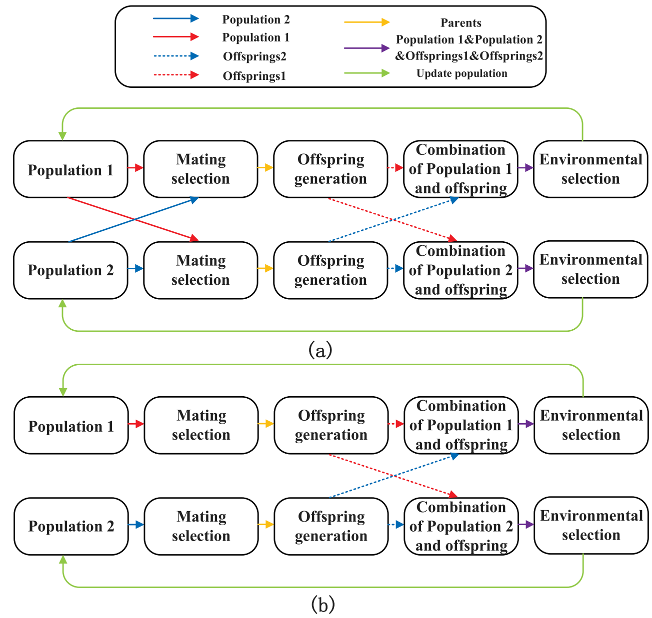

Fig. 4. Two different illustrations of coevolution: (a) strong coevolution, (b) weak coevolution.

proposed a two-archive based constrained multi-objective competitive swarm optimizer (TACSO). The main population aims at feasible solutions, while the auxiliary population aims at convergence. The adaptive δ influences mate selection from the convergence archive. Using strong coevolution, it exploits elite insights to avoid local optima.

(2)Weak coevolution: Each population evolves independently, and does not share population information during the mating selection stage. After generating offspring, the populations are combined with their own offspring and the offspring of other populations, and share all the offspring produced by the populations, as shown in Fig. 4(b). This method maintains population diversity, avoids premature convergence, reduces computational overhead, but it may also underutilize the information between populations, leading to traps in local optima.

Tian et al. [17] first proposed a coevolutionary constrained multi-objective optimization (CCMO) framework, which is flexible and lightweight due to the weak cooperation between populations, and does not require new selection strategies and parameters, thus reducing the computational complexity. Yang et al. [40] proposed a cooperative evolutionary algorithm for dealing with steady-state CMOPs, which differs from the above algorithms [17,35–39,41] in that it finally outputs the auxiliary population. Wang et al. [42] proposed a cooperative multi-objective evolutionary algorithm with propulsive population (CMOEA-PP), where the main population searches for feasible PF and maintains diversity. The auxiliary population initially overlooks constraints, considers them, and searches only for corner and central solutions to speed up convergence in the later stage. Qiao et al. [43] proposed an evolutionary multitasking (EMT)-based constrained multi-objective optimization (EMCMO) algorithm, which determines which populations are useful knowledge based on the four relationships of CPF and UPF in Fig. 1.

Moreover, Ming et al. [44] proposed a tri-population (TriP) based coevolutionary framework, in which the first and second populations target the original problem and the unconstrained problem respectively, and use weak coevolution to handle simple CMOPs; the third population targets the constraint relaxed problem and uses constraint relaxation techniques to evolve. The cooperation of the three populations preserves the advantages of both weak coevolution and constraint relaxation. Li et al. [45] proposed a CMOEA with the two-archive weak cooperation (CMOEA-TWC), which is based on CCMO [17], uses the density-based SDE to enhance the diversity of the auxiliary population, and uses the strict constraint domination principle to improve the feasibility of the main population. Kong et al. [46] proposed a dynamic dual-population coevolution multi-objective evolutionary algorithm (DDCMEA), rooted in CCMO [17]. This method dynamically modulates offspring counts for both populations to balance convergence with feasibility. To boost offspring diversity, genetic and differential evolution algorithms serve as search mechanisms for the auxiliary and main populations, respectively. Cao et al. [47] proposed a coevolutionary constrained multi-objective algorithm with learning constraint boundary, which is based on CCMO [17] and transforms the CMOP into an unconstrained one by taking the constraint violation as an additional objective. The auxiliary population selects solutions according to the learning constraint boundary (LCB). Song et al. [48] proposed a Self-adaptive epsilon constrained multi-objective optimization (SaE-CMO), which is based on CCMO [17], both populations adopt the newly proposed self-adaptive ε method, and use different comparison criteria to select the next population from the mating pool.

2.2.2. Methods based on multi-stage

These methods operate through the sequential application of different strategies or mechanisms at each stage, aiming to progressively refine the solution set towards the PF. Typically, the early stages utilize more information from the objective function, which helps to balance the feasible and infeasible regions and avoid falling into local optima. During the later stages, the focus shifts to information from the constraint conditions, thereby improving the quality of feasible solutions.

Fan et al. [18] first proposed a push and pull search (PPS) framework, which divides the search process into two distinct stages: push and pull search stages. The push stage is the early stage, which can estimate the constraints in CMOPs and thus set the parameters for the constraint handling method in the pull stage; the pull stage is the later stage, which uses an improved ε-constraint handling method to pull the population into the CPF. Jiao et al. [49] proposed a feasible proportion control technique, which can control the proportion of feasible solutions in the population. Different from [18], in the early stage, the feasible ratio control technique is used to handle the constraints, and in the later stage, the commonly used differential evolution algorithm (DE) is used to speed up the convergence. Fan et al. [50] based on [18], combined the multi-objective to multi-objective (M2M) decomposition method with the push and pull search framework (PPS-M2M). Yu et al. [51] proposed a purpose-directed two-phase multi-objective differential evolution (PDTP-MDE) algorithm. Tian et al. [52] proposed a two-phase evolutionary algorithm that can switch between two phases according to the current population state. Yu et al. [53] proposed a dynamic selection preference-assisted constrained multi-objective differential evolutionary algorithm. Ming et al. [54] proposed a two-stage framework to improve the efficiency and generality of CMOPs. Xu et al. [55] proposed a CMOEA (C-TPEA) with two-stage constraint handling. In the early stage, the grid constraint decomposition method framework can well maintain the convergence and diversity; in the later stage, the ε-constraint strategy is introduced to further improve the feasibility and quality of the solution. Ma et al. [56] proposed a strategy to rank the constraint handling priorities based on the impact on the unconstrained PF, and added the constraint conditions one by one according to the constraint handling priority, and processed them at different stages of evolution.

Moreover, Jiao et al. [57] proposed two-types weight adjustments based on decomposition-based multi-objective evolutionary algorithm (MOEA/D). Feng et al. [58] proposed a three-phase hybrid driving strategy for selecting mating parents. In the first phase, tournament selection for feasibility is used to select mating parents. In the second phase, tournament selection for diversity is used to select mating parents. During the third phase, the population is distributed in the proximity of the PF. The main objective of this phase is to further optimize the population towards the PF while maintaining stability. Tournament selection for feasibility is utilized to select mating parents. Zhang et al. [59] proposed a non-parametric technique for handling constraints in the later stage. This approach makes full use of the hidden information in the feasible non-dominated set. Sun et al. [60] addressed the complex multi-constraint situation. In the early stage, they obtained the priority among the constraints based on the relationship between the PF of each constraint and their common PF. In the middle stage, they evaluated each constraint according to the priority and further explored the objective space. In the late stage, they fully considered the feasibility, improved the quality of the solutions obtained in the previous two stages, and finally obtained solutions with good convergence, feasibility, and diversity. Song et al. [61] proposed a two-stage differential evolution algorithm (TSCMODE). Liang et al. [62] explored and exploited the relationship between CPF and UPF to solve CMOPs, taking the early stage as the learning stage, measuring the relationship between CPF and UPF; the later stage as the evolutionary stage, calculating specific evolutionary strategies based on the learned relationship, to improve the utilization efficiency of objective information.

Besides, Liu et al. [63] aimed at expensive multi-objective constrained optimization problems, proposed a surrogate-assisted two-stage differential evolution (SA-TSDE). Zou et al. [64] aimed at complex feasible regions, proposed a more flexible two-stage evolutionary algorithm based on automatic regulation (ARCMO), which performs fast global search in the early stage, and passes the population information to the later stage. The later stage consists of two dynamically complementary subprocesses: exploration subprocess and convergence subprocess. Xia et al. [65] proposed a novel CMOEA with two-stage resources allocation, which dynamically allocates resources to two stages. Feng et al. [66] proposed an adaptive tradeoff evolutionary algorithm (ATEA), which divides the search process into three stages. In the early stage, it guides the constraint relaxation technique to perform global search, making the population quickly cross the infeasible region and approach the feasible region. In the middle stage, it further detects and estimates the constraint conditions, to retain more feasible individuals and competitive infeasible individuals, thus making the population accurately identify all possible feasible regions, and gradually extend to the feasible boundary. In the later stage, it explores the areas that are insufficiently explored in the previous stages, accelerates the population convergence, and escapes from the local optimum state. Chen et al. [67] aimed at dynamic constrained multi-objective optimization problems, proposed a two-stage diversity compensation strategy. Huang et al. [68] proposed a framework for a CMOEA that utilizes global and local FS searches. The algorithm uses a global search operator in the early stage and a local search operator in the middle stage. The first two stages are not subject to constraint restrictions, and the feasible solutions are stored in the FeasiblePool for environmental selection. In the later stage, the information regarding these feasible solutions is reused to guide the population evolution while considering various constraint conditions.

2.2.3. Methods based on chaotic optimization

These methods typically use chaotic maps to generate pseudo-random numbers that are mapped as decision variables for global optimization problems [69]. They are usually combined with a genetic algorithm to increase the diversity of the population through chaotic initialization, or adjust the algorithm parameters through chaotic mapping, so as to improve the search efficiency and solution quality of the algorithm.

Aslimani et al. [70] proposed a new method based on chaotic search. The method is characterized by fast and accurate convergence, as well as parallel and independent decomposition of the target space. Li et al. [71] proposed the parallel chaotic optimization algorithm (PCOA) as an improvement to the chaotic optimization algorithm (COA). PCOA is a population-based optimization algorithm that reduces sensitivity to initial values. Zhang et al. [72] proposed a multi-objective evolutionary algorithm (LCMO) based on a weakly cooperative evolutionary framework and multiple chaos operators. The hybrid chaos operator is used to improve the distribution uniformity of candidate solution populations, and the quadratic selection criterion refines the Pareto level. Takubo and Kanazaki [73] proposed an integrated optimization method based on CMOEA and non-intrusive polynomial chaos expansion (PCE) for solving robust constrained multi-objective optimization problems under time-series dynamics. Yacoubi et al. [74] proposed a multi-objective chaos game optimization algorithm (MOCGO/DR), by embedding theoretical concepts such as multi-objective, decomposition, and stochastic learning in the traditional CGO algorithm. Rauf et al. [75] proposed the multiple population-based chaotic differential evolution (MPC-DE) algorithm. The algorithm divides the population into two sub-populations using an enhanced chaos-based population initialization method. The first sub-population uses a sinusoidal chaotic initialization method, while the second sub-population uses a tent-mapped chaotic initialization method. Each sub-population adopts an improved mutation strategy based on a two-dimensional chaotic mapping. The proposed mutation strategy generates mutation vectors, and a corresponding selection criterion is used to select the optimal progeny generated by each sub-population.

In general, methods based on evolution are known for their simplicity, flexibility, and minimal parameter requirements, effectively balancing constraints and goals while preserving population diversity. However, they can incur high computational costs, especially with archived populations during co-evolution, and may struggle with constraint-objective balance, causing poor convergence. Additionally, reliance on prior knowledge when addressing constraints may hinder performance in complex scenarios.

2.3. Methods based on machine learning

These methods address optimization challenges through a data-driven approach, capable of learning from historical data to predict outcomes. They are aptly suited for dynamic settings and online optimization issues. The methods are categorized into three distinct types: algorithm recommendation, intelligent parameter tuning, and predictive modeling.

2.3.1. Algorithm recommendation

Tian et al. [76] proposed an automated algorithm selection method for choosing the most suitable evolutionary algorithm for a given MOP, where the input is the explicit and implicit features of multi-objective optimization problems sampled by Latin hypercube, and the output is the evolutionary algorithm with the best performance on multi-objective optimization problems. Qiao et al. [77] proposed the first evolution-based constrained multi-objective feature extraction method (ECMOFE), where two populations target the constraints and objectives respectively, and adopt two complementary evolutionary strategies to co-evolve. In the evolutionary process, the feature matrix is obtained by recording the success rate of the offspring population. Then, a dimensionality reduction method is designed to reduce the size of the matrix without losing important information, and then the matrix is transformed into a vector, and then trained by a classifier. Han et al. [78] proposed a multi-heuristic algorithm based on deep reinforcement learning, which dynamically selects the most suitable meta-heuristic algorithm (discrete cuckoo search and particle swarm optimization) based on the diversity and quality of the population during the population evolution process, and designs rewards based on the evolution effect, thereby guiding the algorithm to update the solutions more effectively and achieve global optimization.

Moreover, Zuo et al. [19] proposed the process knowledge-guided constrained multi-objective autonomous evolutionary optimization method, which is different from [76,77] in that it does not recommend CMOEAs, but intelligently recommends the subsequent guidance strategies according to the established mapping model and the current population state, and adopts appropriate strategies at different stages of population evolution, without changing the structure of the existing evolutionary algorithm, but using process knowledge to guide the population evolution. A multilayer perceptron is used to train the mapping model. Wang et al. [79] proposed a transfer-based method to enrich the algorithm library for constrained multi-objective optimization problems. This method calculates the similarity between problems based on the landscape features of the problems, and then calculates the transfer probability of intelligent optimization algorithms that solve similar problems based on the performance of each algorithm and the similarity between problems. Algorithms with high transfer probability will enrich the algorithm library for solving the current problem. This method uses a Softmax regression model to recommend algorithms for solving the current problem. Ming et al. [80] proposed an adaptive selection of auxiliary tasks for multi-task auxiliary constrained multi-objective optimization. The method transforms CMOPs into a multi-task optimization problem. The original CMOPs serve as the main task, while several auxiliary tasks use different algorithmic strategies. The method uses reinforcement learning and deep learning techniques to dynamically select the most suitable auxiliary tasks based on the current state of the population, thereby improving the optimization results.

2.3.2. Intelligent parameter adjustment

Fallahi et al. [81] proposed using a reinforcement learning algorithm, Q-learning, to intelligently tune the parameter values of DE and particle swarm optimization (PSO) algorithms. They also developed two new variants of DE and PSO, namely DE with Q-learning (DEQL) and PSO with Q-learning (PSOQL). Yu et al. [82] proposed the reinforcement learning-based multi-objective differential evolution (RLMODE) algorithm. During the evolution process, the offspring generated is evaluated and compared with its corresponding parents, the relationship between the offspring and parent can adjust the parameters of RLMODE by the reinforcement learning (RL) technique. The feedback mechanism can generate the most suitable parameters for RLMODE, thus pushing the population towards the feasible region.

2.3.3. Prediction generation

Ma et al. [83] proposed an adaptive reference vector reinforcement learning method based on decomposition algorithms, which aims to solve the problem of decomposition-based algorithms being sensitive to the shape of the PF. In this method, the adaptation process of the reference vectors is regarded as a reinforcement learning task, where each reference vector can learn from the environmental feedback and choose the actions that best fit the problem characteristics. Moreover, the sampling operation of the reference points uses a distribution estimation learning model to generate new reference points. Wang et al. [84] proposed a model-based evolutionary constrained multi-objective optimization offspring generation strategy. This strategy uses a growing neural gas network to collect feasibility information during the evolutionary search process, and gradually builds a distribution model of the feasible region in the decision space. In order to make full use of infeasible solutions to assist the modeling of the feasible region, this strategy treats feasible solutions and infeasible solutions as two classes of samples, and uses supervised learning methods to train the network. It can use the learned network to generate promising offspring, thereby accelerating the convergence of the population to the optimal feasible region.

Moreover, Ming et al. [20] proposed a new machine learning-based method, which adopts the determinantal point process to select the population and archive, instead of the traditional distance- or density-based selection strategies. The determinantal point process is a probabilistic model that can quantify the diversity of a subset, and it can effectively extract representative and balanced solutions from a

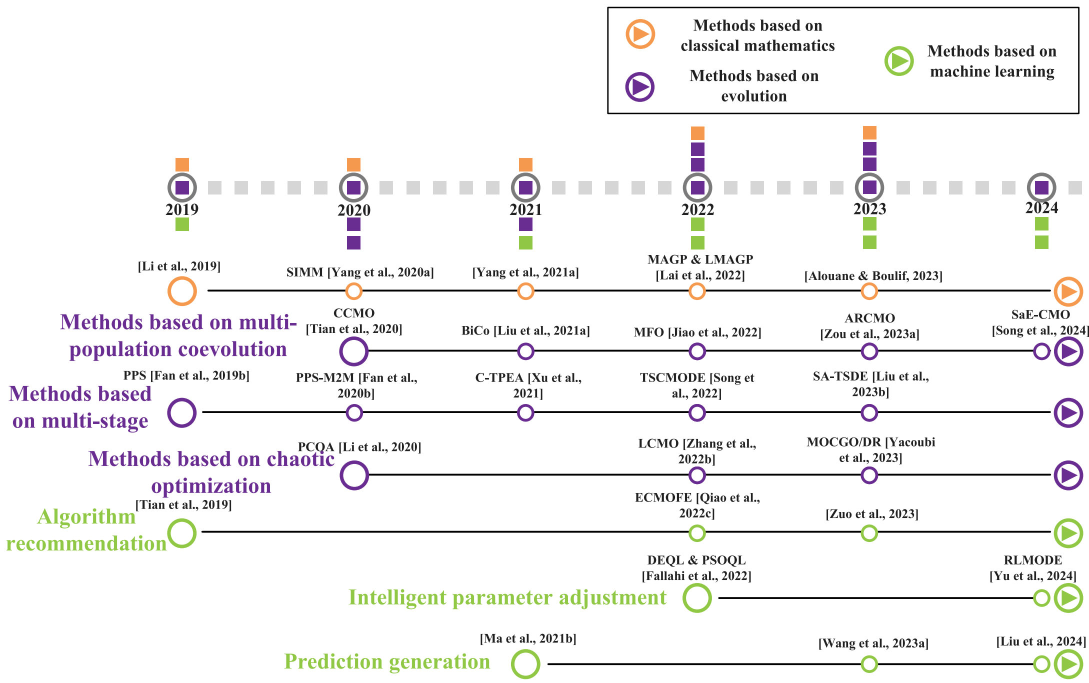

Fig. 5. The development of current CMOAs: a selection of representative algorithms.

candidate set [85] . To apply the determinantal point process in multi-objective optimization, the method designed two innovative kernel matrices, which reflect the convergence, diversity, and feasibility of the solutions in the population and archive, respectively, thus achieving efficient exploration and exploitation of the solution space. Based on [20], Hu et al. [86] proposed a reinforcement learning-based population state evaluation method, which uses the overall improvement degree as the reward signal to guide the neural network to select the appropriate parent population and archive for mutation operations. In addition, this method also introduced the idea of coevolution into the generation of training data, by sharing information among multiple populations, enhancing the diversity of the samples, and thus improving the accuracy of the neural network model. Liu et al. [87] proposed a CMOEA with Pareto estimation by neural network, using a dual population mechanism, and training the network with self-organizing map (SOM). The algorithm preserves neighborhood information, maps the population structure of the decision space to the objective space, and then uses neuron weights to estimate the PF.

In general, machine learning-based methods excel in adapting to complex environments and enhancing algorithm robustness, offering great promise. Yet, their performance hinges on feature extraction, demanding extensive data and computational power, which could escalate algorithm complexity and diminish interpretability.

2.4. Discussion

The strengths and weaknesses of the above categories of existing CMOAs will be discussed here mainly and summarized in Table 1. Additionally, Fig. 5 displays a timeline of the development of current CMOAs.

(1) Methods based on classical mathematics: In methods based on classical mathematics, solving constrained multi-objective optimization problems usually involves transforming multiple objectives into a single objective to leverage established optimization algorithms. These methods have a solid theoretical foundation, a clear computational procedure, and often converge quickly to a solution. Linear programming is a widely used technique that finds the optimal solution by linearizing objectives and constraints. The ϵ-constraint method selects one main objective from multiple objectives to optimize, while the other objectives are used as constraints. Fuzzy set theory, on the other hand, uses fuzzy logic to deal with uncertainty, allowing solutions to be elastic in the degree of satisfaction. However, it is important to note that these methods have limitations. First, multi-objective problems are often reduced to single-objective problems, which can result in misunderstandings or oversimplifications of the original problem. Additionally, determining the weight or ϵ value may require subjective judgment from the decision maker, which can make it difficult to ensure the quality of the solution. In practical applications, these methods may require more tuning and adaptation to obtain an effective solution, as they may not effectively handle nonlinear or non-convex problems due to their reliance on the linear and convex nature of the problem to ensure solution quality and algorithm convergence.

(2) Methods based on evolution: Methods based on evolution are more flexible than classical mathematics optimization methods and can handle nonlinear, non-convex, or non-differentiable problems. This is because evolution methods do not rely on the mathematical properties of the problem, but approach the optimal solution through iterative search. However, this approach also has its limitations. As trial-and-error based methods, they may require significant computational resources to find satisfactory solutions. This is particularly noticeable in problems with high dimensions or complex constraints, where the search space is vast, and the algorithm requires more time to explore. Additionally, the performance of the evolutionary methods is highly dependent on the proper setting of its parameters. If the parameters are not selected appropriately, the algorithm may fail to converge to the optimal solution, or the convergence rate may be very slow. The algorithm designer must possess a thorough understanding of the problem and extensive experience in parameter tuning.

(3) Methods based on machine learning: According to the no free lunch theorem [88], any algorithm's performance advantage on one class of problems becomes a disadvantage on another class of problems. Based on machine learning methods, this approach can use data and models to automatically learn and adapt to complex problem environments, improving the algorithm's generalization and robustness. This approach can construct nonlinear mappings and strategies through deep learning or reinforcement learning methods, breaking through the limitations of traditional algorithms and improving the algorithm's flexibility and innovation. Feature extraction, which is essential for training

Table 1
Advantages and limitations of various methods.

Categories	Advantages	Limitations
Methods based on classical mathematics	The methods in solving CMOPs include a solid theoretical foundation, clear computational procedures, and the ability to converge quickly to a solution.	These methods may lead to misinterpretation or oversimplification of the original problem, and may also require subjective judgment to determine weights or values, which can affect the quality of the solution, and may not be able to deal effectively with nonlinear or nonconvex problems.
Methods based on evolution	The methods are more flexible, can handle nonlinear, non-convex, or non-differentiable problems, and do not rely on the mathematical nature of the problem.	They may demand extensive computational resources and sensitive parameter tuning to effectively converge on an optimal solution.
Methods based on machine learning	The methods uses data and models to automatically learn and adapt to complex problem environments, improving the generalization and robustness of the algorithms.	Feature extraction directly affects the performance of the algorithm, and also requires a large amount of data and computational resources, which may make the algorithm more complex in time and space and less interpretable.

models, directly affects the algorithm's performance. In addition, model training requires a lot of data and computing resources, which may increase the algorithm's time and space complexity and make it difficult to achieve real-time and online optimization. Moreover, model design and parameter setting need to be reasonable, otherwise they may reduce the algorithm's stability and interpretability and make it difficult to guarantee the algorithm's reliability and controllability.

Despite the excellent performance of existing algorithms in solving CMOPs, some research gaps still need to be considered:

- With the development of deep learning, large language models have achieved remarkable results in the field of natural language processing. Large language models can be pre-trained on large-scale text data, learning rich language knowledge, and then complete different downstream tasks by fine-tuning or zero-shot learning. Liu et al. [89] combined large language models with multi-objective optimization and used suitable prompt engineering to make the general large language models act as black-box search operators for MOEA/D in a zero-shot learning manner. They showed powerful performance. Their method can effectively handle the balance between constraints and objectives in multi-objective optimization problems and improve the search efficiency and solution quality by using the powerful expression of large language models. However, research on the combination of constrained multi-objective optimization and large language models is still lacking. We think this direction is challenging and promising, and also a new development trend of constrained multi-objective optimization in the future. Large language models can adapt to different problem types by using different prompt engineering as a general search operator, achieving an effective balance between constraints and objectives. This direction of research cannot only expand the application scope of large language models, but also provide a new idea and method for solving constrained multi-objective optimization problems.
- In this section, we mentioned the machine learning-based method that uses a data-driven approach to solve CMOPs. However, the performance of the machine learning-based method depends largely on the effect of feature extraction, which is the process of transforming raw data into meaningful information. The machine learning-based method will fail or be inefficient if feature extraction cannot fully reflect the data's characteristics and the complexity of the CMOP. Therefore, feature engineering research is an urgent topic for CMOPs. The purpose of feature engineering research is to find an effective feature extraction method that can extract features from the original data, which are helpful for optimizing CMOPs, and reduce the dimensionality and complexity of the data, improving the computational efficiency and interpretability.
- Some methods based on machine learning use supervised learning to train mapping models, but a general dataset for researchers to conduct experimental exploration is still lacking. Due to their diversity and complexity, CMOPs make it difficult to find a general dataset that can cover different problem characteristics and scenarios. Researchers usually use some artificially constructed test functions or some datasets from practical applications to conduct experiments, but these datasets often fail to reflect the reality and universality of CMOPs. Therefore, a meaningful research direction is to design a general dataset that can simulate different aspects of CMOPs, such as the type, number, difficulty, and dynamics of constraints. Such a dataset can provide researchers with a fair and comprehensive evaluation platform and promote the development and improvement of machine learning-based methods.

3. Applications

This section presents the applications of CMOAs in some representative domains, such as engineering design, resource optimization, scheduling optimization, and path planning. These domains involve problems that are of practical significance and value. Table 2 provides detailed information on the practical application of some CMOAs.

3.1. Engineering design problems

Engineering design problems are those that can be optimized or improved in engineering practice. The objective is to identify the optimal or near-optimal design solutions that meet certain performance, safety, economic, and relevant requirements. It mainly includes gearbox design [128], heat exchanger network design [129], disc brake design [130], spring design problems [131], twin-rod truss design [132], car side impact problem [133], reducer design problem [134], the simply supported I-beam design problem [135], bulk carrier design [136], pressure vessel design [137], reactor network design [138], vibration platform design problem [90], the process synthesis and design problem [92], software engineering [93], welded beam design [95], antenna design optimization [98], robot gripper optimization [99], optimal wireless sensor location design [1], dual source compressed air piping optimization problem [100], servo turntable system [103], turning process parameters optimization [139], Al-Li alloy thin-wall workpieces design [140].

Kohli and Arora [97] introduced chaos theory into the GWO algorithm with the aim of accelerating its global convergence speed to solve the spring design problem, gear train design problem, welded beam design problem, pressure vessel design problem. Ming et al. [109] proposed a novel constrained multi-objective optimization evolutionary algorithm with enhanced mating and environmental selections (CMME) for gearbox design and heat exchanger network design problems. Zou et al. [125] proposed a dual-population constrained optimization algorithm based on surrogate evolution and degeneration (CAEAD) to solve the heat exchanger network design problem. Kong et al. [46] proposed a dynamic dual-population cooperative evolutionary multi-objective evolutionary algorithm to solve the disc brake design problem and

Table 2
Summary of practical applications of CMOAs.

Scenarios	Problems	Algorithms	The number of objectives	The number of constraints
Engineering design problems	vibration platform design problem [90]	AT-CMOEA [91], CMOEA-MS [52]	2	5
	the process synthesis and design problem [92]	AT-CMOEA [91]	2	2
	software engineering [93]	ToM [94]	2,3,4	20
	welded beam design [95]	PPTA [96], GLS-CMOEA [68], [97]	2	3
	antenna design optimization [98]	MOEA/D-2WA [57]	3	2
	robot gripper optimization [99]	PPS [18]	2	8
	optimal wireless sensor location design [1]	DCCMO [1]	2	2,9
	dual source compressed air piping optimization problem [100]	MODE-CHS [101], DDCMEA [46], MODE-PS [102]	2	4
	servo turntable system [103]	MOPSO [103]	4	3
Resource optimization problems	water distribution network optimization [104]	C-TAEA [39]	4	2
	groundwater remediation design [105]	AMALGAM [105]	2	4
	thermocline filling layer thermal energy storage [106]	[106]	2	3
	test resource allocation optimization [107]	ECHTs [107]	3	3
	combined cooling heating and power (CCHP) microgrids optimization [29]	MILP [29]	2	17
	economic load dispatch problem (EDP) [75]	MPC-DE [75]	2	8
Scheduling optimization problems	multi-product batch plant problem [108]	CMME [109], ARCMO [64], AT-CMOEA [91]	3	3
	dual-resource constrained flexible job shop scheduling problem [110]	TS-MOPSO [110]	3	19
	short-term crude oil scheduling problem [111]	NSGAII-APE [111]	5	10
	urban bus scheduling problem [112]	ShiP [113]	2	5
	combined economic emission dispatch (CEED) problem [114]	CMOPEO-EED [115], [116], [117]	2	9
	optimal reactive power dispatch (ORPD) problem [118]	[118]	2	5
Path planning problems	unmanned aerial vehicle (UAV) planning [119]	M2M-DW [119]	2	4
	path planning for emergency material scheduling [120]	[120]	3	8
	fuzzy business restricted multi-objective multi-index transportation problem [121]	[121]	2	3
	robot trajectory planning [122]	MOGA [122]	2	5
	constrained multi-objective solid traveling salesman problems [123]	R-MOGA [123]	2	1
Other problems	process synthesis problems [124]	CAEAD [125], MCCMO [126]	2	9
	polyester polymerization process problems [127]	IARVEA [127]	4	3
	gasoline blending process problem [30]	MOEA/D-FCHT [30]	2	10

the dual-source compressed air pipeline optimization problem. Yang et al. [141] proposed a novel CMOEA assisted by an additional objective function (CMAOO) to solve the disc brake problem. Gu et al. [142] proposed an improved constrained dominance principle (ICDP) embedded in MOEA/D (MOEA/D-ICDP) to solve the spring design problem and the disc brake problem. Yuan et al. [143] proposed a criterion-based constrained evolutionary algorithm (CCEA) for evaluating the value of infeasible solutions to solve the twin-rod truss design problem. Xia et al. [65] proposed a novel two-stage resource allocation CMOEA (CMOEA-TSRA) to solve the car side impact problem, the reducer design problem and the disc brake problem. Zou et al. [64] proposed a more flexible two-stage evolutionary algorithm based on automatic adjustment to solve the disc brake design problem, the simply supported I-beam design problem, the gearbox design problem, and the car side impact problem.

Moreover, Yang et al. [101] proposed a multi-objective differential evolution algorithm based on dominance and constraint handling switching mechanism (MODE-CHS) to solve the disc brake design problem and the dual-source compressed air pipeline optimization problem. Sun et al. [60] proposed a multi-stage CMOEA with constraint condition priority (C3M) to solve the bulk carrier design, pressure vessel design, and reactor network design problems. Ma et al. [56] proposed a multi-stage evolutionary algorithm with stepwise addition of constraints (MSCMO) to solve the reducer design problem, the disc brake design problem, and the car side impact problem. Bao et al. [91] proposed an archive-based two-stage evolutionary algorithm (AT-CMOEA) to solve the vibration platform design problem and the process synthesis and design problem. Xiang et al. [94] proposed a tradeoff model (ToM) to solve the constrained optimal software product selection problem in search-based software engineering. Tian et al. [52] proposed CMOEA with a two-stage framework (CMOEA-MS) to solve the car side impact problem and the vibration platform design problem. To effectively handle CMOPs with complex infeasible regions, Qin et al. [96] proposed a dual-archive assisted push-pull evolutionary algorithm (PPTA) to solve the welded beam design problem. Huang et al. [68] proposed a CMOEA framework based on global and local FS search (GLS-CMOEA) to solve the reducer design problem, the welded beam design problem, and the disc brake design problem. To solve high-constrained multi-objective optimization problems, Jiao et al. [57] proposed two types of weight adjustment methods based on MOEA/D (MOEA/D-2WA) to solve the antenna design optimization problem.

Besides, Song et al. [48] proposed an adaptive ε-constraint multi-objective optimization and adopted weak cooperative evolution, to solve the bulk carrier design problem and the reactor network design problem. Jiao et al. [37] proposed a multi-task optimization framework to solve the antenna array synthesis problem. Fan et al. [18] proposed an improved ε-constraint handling push-pull search framework for robot gripper optimization problem. Yu et al. [1] proposed a dual-population constrained multi-objective optimization algorithm (DCCMO) to find the optimal wireless sensor location. Yang et al. [102] proposed a multi-objective differential evolution algorithm based on partition selection (MODE-PS), to solve the dual-source compressed air pipeline optimization problem. Zhang et al. [103] proposed a switching nonlinear autoregressive moving average model (MOPSO) for continuous electromechanical servo turntable system, and developed a multi-objective particle swarm optimization algorithm, which implements the global best position selection of each particle and the maintenance of the external archive in the same program, to optimize the performance of the servo turntable system. Gadagi and Adake [139] considered the process parameters such as spindle speed, cutting depth and feed rate, and performed turning on glass fiber reinforced composite materials, and optimized the turning process parameters by genetic algorithm and particle swarm optimization techniques, to improve the turning quality and efficiency. Yue et al. [140] studied the impact of milling parameters on surface roughness and specific cutting energy. They developed a multi-objective optimization model and employed a multi-objective particle swarm optimization algorithm with a chaotic local search and adaptive inertia weights to find the best milling parameter combination.

3.2. Resource optimization problems

As the concept of environmental protection becomes more popular, resource optimization problems have attracted widespread attention. Effectively utilizing resources, reducing resource waste, and minimizing environmental pollution are the objectives of a constrained multi-objective optimization problem. It mainly includes water resources optimization [144], green mix design of rubberized concrete [145], integrated coal mine energy system [146], wastewater control problem [147], hybrid renewable energy systems [148], wind farm layout optimization [2], water resource management problem [149], water distribution network optimization [104], groundwater remediation design [105], thermocline filling layer thermal energy storage [106], test resource allocation optimization [107], CCHP microgrids optimization [29], EDP [75].

Tian et al. [52] proposed a two-stage evolutionary algorithm that adapts the fitness assessment strategy during the evolutionary process to balance objective optimization and constraint satisfaction for solving water resource optimization problems. Golafshani et al. [145] used machine learning ensemble models and multi-objective gray wolf optimizer to estimate the composition of rubber concrete and achieve its green design. Wang et al. [79] proposed a migration-based method of enriching the algorithm library. This method selects and adds suitable algorithms to the library based on problem similarity and algorithm performance, and then uses Softmax regression to create an optimal algorithm combination that solves the coal mine comprehensive energy system optimization problem. Hou et al. [150] proposed a multi-state constrained multi-objective differential evolution with variable neighborhood strategy (MSCMODE-VNS) for wastewater control optimization. Ming et al. [148] designed a CMOEA based on two-stage and two-population (DD-CMOEA) for hybrid renewable energy system (HRES) optimization. Yin et al. [151] developed a code design optimization framework, jointly optimizing the wind farm layout and pumped storage power station configuration, and constructed an adaptive neuro-fuzzy inference system (ANFIS) trained by evolution to predict the power output of offshore wind farms. Qin et al. [96] proposed a dual-archive assisted push-pull evolutionary algorithm to solve the water resources management problem.

Moreover, Li et al. [39] proposed a parameter-free dual-archive evolutionary algorithm (C-TAEA), and developed a constrained mating selection mechanism, to solve the water distribution network optimization problem. Ouyang et al. [105] proposed a multi-algorithm gene adaptive multi-objective method (AMALGAM), as a multi-objective optimization solver, by incorporating the uncertainty of surrogate modeling into the optimization model through chance-constrained programming (CCP), to solve the groundwater remediation design problem of high-density non-aqueous phase liquid contaminated sites. Marti et al. [106] used constrained multi-objective optimization method to optimize the energy efficiency and material cost of a packed bed thermal energy storage system with air as the heat transfer fluid. Su et al. [107] proposed a multi-objective test resource allocation problem model with preset reliability, and developed several enhanced constraint handling techniques (ECHTs) derived from the new lower bound, to correct and reduce the violation of constraints, to solve the test resource allocation optimization problem. Yang et al. [29] developed a model for multi-objective optimal dispatch in CHP microgrids. The model integrates renewable energy sources, energy storage systems, and incentive demand response to enable economic optimization and peak load reduction. The problem was solved using an mixed-integer linear programming (MILP) based on the extended ε-constraints approach. Rauf et al. [75] proposed a multiple swarm-based chaotic differential evolutionary algorithm (MPC-DE) for the EDP.

3.3. Scheduling optimization problems

Scheduling optimization involves efficiently organizing tasks or activities within constrained time and resources to meet specific objectives. It mainly includes multi-product batch plant problem [108], dual-resource constrained flexible job shop scheduling problem [110], short-term crude oil scheduling problem [111], urban bus scheduling problem [112], CEED problem [114], ORPD problem [118].

Ming et al. [109] proposed a novel CMOEA, which improved the mechanisms of crossover and environment selection, adopted two innovative ranking strategies and a novel individual density estimation method, thus effectively solving the multi-product mixing plant problem. Zou et al. [64] proposed a more flexible two-phase evolutionary algorithm based on adaptive regulation, thus effectively solving the multi-product batch plant problem. Bao et al. [91] effectively solved the multi-product batch plant problem by proposing an archive-based two-phase evolutionary algorithm. Shi et al. [110] considered the problem of increased worker boredom caused by repetitive work assignments, and constructed an efficiency function to characterize the impact of worker boredom, established a two-layer dictionary model, with effectively completing all production tasks as the primary optimization objective, and improving worker job satisfaction as the secondary optimization objective, proposed a two-phase multi-objective particle swarm optimization algorithm that adopts a three-dimensional representation scheme (TS-MOPSO), to solve the dual-resource constrained flexible job shop scheduling problem. Hou et al. [111] proposed an adaptive enhanced selection pressure algorithm based on NSGA-II-CDP (NSGAII-APE), which could effectively enhance the selection pressure in the later iterations, thus effectively solving the problem of simultaneous processing of low-melting-point oil and high-melting-point oil.

Moreover, Ma and Wang [113] proposed a constraint handling technique called shift-based penalty (ShiP) to solve the vehicle scheduling problem of urban bus routes. Chen et al. [115] proposed a constrained multi-objective population extremal optimization algorithm named CMOPEO-EED, which found the Pareto optimal solutions of the renewable energy multi-objective economic emission dispatch problem by minimizing the total cost and environmental emission, two conflicting objectives. El-Shorbagy and Mousa [116] proposed a new optimization system named constrained multi-objective equilibrium optimizer, which combined the dominance criterion to handle the multi-objective functions, and reduced the size of the PF to a reasonable one by clustering analysis, applied repair methods to handle the particles' constraint conditions and feasibility, thus better solving the CEED problem. Yu et al. [117] designed a new DE mutation strategy and Gaussian mutation to maintain the convergence and diversity of feasible solutions, thus solving the CEED problem. Mohseni-Bonab et al. [118] studied the stochastic multi-objective ORPD (SMO-ORPD) problem in wind-integrated power systems, considering the uncertainty of load and wind power generation, and used the ε-constraint method and fuzzy satisfaction method to select the best trade-off solution.

3.4. Path planning problems

Path planning involves identifying the most appropriate route from start to finish, considering various constraints and context-specific criteria such as distance, speed, safety, and energy efficiency. It mainly includes UAV planning [119], vehicle routing problems with time windows [152], path planning for emergency material scheduling [120], fuzzy business restricted multi-objective multi-index transportation problem [121], UAV path planning problem for disaster relief [82], robot trajectory planning [122], constrained multi-objective solid traveling salesman problems [123].

Peng and Qiu [119] proposed a decomposition-based constrained multi-objective evolutionary algorithm (M2M-DW) with a local infeasibility utilization mechanism for UAV path planning. Hou et al. [150] proposed a multi-state constrained multi-objective differential evolution algorithm with a variable neighborhood strategy to improve the search efficiency of complex feasible regions, which was used to solve the multi-objective vehicle routing problem with time windows. Tian et al. [17] proposed a weakly cooperative coevolutionary framework to improve the solving ability for the multi-objective vehicle routing problem with time windows through bi-directional information exchange. Wang et al. [120] established a drone-based emergency medical service queue location model, which used fuzzy theory to deal with the fuzziness of drone endurance time and demand arrival rate under the priority queuing strategy, and used multi-objective optimization methods to balance the total cost, system efficiency and fair response time, effectively solving the emergency medical service path planning problem. Latpate and Kurade [121] proposed a two-stage fuzzy business constrained multi-objective multi-index transportation problem, where the supply, demand and requirement were triangular fuzzy numbers, and adopted a fuzzy non-dominated sorting genetic algorithm, which used the alpha-cut technique to deal with the uncertainty of the problem parameters, and performed the frontier non-dominance sorting of the parent population according to the uncertainty level, effectively solving the fuzzy business constrained multi-objective multi-index transportation problem. Yu et al. [82] proposed a reinforcement learning-based multi-objective differential evolution algorithm, which can dynamically adjust the main parameters of the algorithm through reinforcement learning techniques, effectively solving the disaster relief drone path planning problem. Chen and Pham [122] proposed a technique based on constrained multi-objective genetic algorithm (MOGA) for solving the general electric motor-driven parallel manipulator problem. Maity et al. [123] proposed a rough multi-objective genetic algorithm (R-MOGA) for solving the constrained multi-objective solid traveling salesman problem in rough, fuzzy rough and stochastic rough environments.

3.5. Other problems

Here, we discuss some problems that do not belong to any of the previous categories. These include process synthesis problems [124], synchronous optimized pulse-width modulation problem for three-stage inverters [153–155], synchronous optimized pulse-width modulation problem for five-stage inverters [153–155], synchronous optimized pulse-width modulation problem for thirteen-stage inverters [153–155], polyester polymerization process problems [127], gasoline blending process problem [30], portfolio optimization [3], agricultural planting structure optimization [156], capacitated multi-facility location problem [157], the landing trajectory design of supersonic transport (SST) [73].

Zou et al. [125] proposed a dual-population constrained optimization algorithm based on the idea of surrogate evolution and degeneration, and can effectively solve the process synthesis problem and the synchronous optimization pulse width modulation problem of five-level and three-level inverters. Zou et al. [126] proposed a new multi-population CMOEA (MCCMO) that uses two new techniques: activation dormancy detection and combine occasion detection (COD). These techniques accelerate the optimization process and determine the best time to find SubCPF. As a result, they effectively solve the process synthesis problem and the synchronous optimization pulse width modulation problem of five-level and thirteen-level inverters. Zhu et al. [127] aimed at the four-objective optimization problem in the polyesterification process, proposed an interactive adaptive reference vector guided multi-objective optimization evolutionary algorithm (IARVEA). Han et al. [30] proposed a fuzzy constraint handling technique, which uses fuzzy set theory to accurately characterize the difference of solutions in objective function values and constraint violation degrees, and embeds it into MOEA/D (MOEA/D-FCHT), thus effectively solving the gasoline blending process problem.

Moreover, Lwin et al. [3] extended the Markowitz mean–variance portfolio optimization model, and proposed an efficient learning-guided hybrid multi-objective evolutionary algorithm. This algorithm effectively solves the constrained portfolio optimization problem under the extended mean–variance framework. Peykani et al. [158] proposed a novel Fuzzy MultiPeriod Multi-Objective Portfolio Optimization (FMPMOPO) model that is capable to be used under data ambiguity and practical constraints including budget constraint, cardinality constraint, and bound constraint, and thus is applicable to the presence of uncertain data in portfolio optimization. Yang et al. [156] utilized the SIMM method to handle fuzzy numbers in the constraints. They then introduced a linear objective planning method to solve the problem of fractional objectives as fuzzy objectives, in order to optimize the agricultural planting structure. Ozgen and Gulsun [157] addressed the multi-objective, capacitated, multi-facility location problem by integrating multiple perspectives to evaluate quantitative and qualitative factors and by combining a two-stage probabilistic linear programming approach with fuzzy hierarchical analysis. Takubo and Kanazaki [73] proposed an integrated optimization method based on CMOEA and non-intrusive PCE for the landing trajectory design of supersonic transport with wind uncertainty.

4. Future research directions

CMOAs have achieved success on multiple CMOPs, but there are still several research directions to explore in the field of constrained multi-objective optimization. This section introduces the important topics in this area.

4.1. Feature extraction based on constrained multi-objective optimization

Section 3 discussed how feature extraction affects the performance of the machine learning-based methods for CMOPs. Feature extraction is the process of extracting information that reflects the essence and characteristics of a problem from the original data. [19,76,77] and some other studies have proposed methods for feature extraction, but feature extraction for CMOPs still faces these difficulties:

- The feature space of CMOPs is usually high-dimensional, containing various information such as objective functions, constraints, and decision variables. Such a feature space makes the computational complexity of feature extraction high and increases the difficulty of feature selection.
- The features of CMOPs often have correlation or redundancy, affecting the effectiveness and stability of feature extraction. The correlation between features indicates a certain statistical relationship between features, for example, linear correlation, nonlinear correlation, or monotonic correlation.
- The features of CMOPs often have nonlinear or non-convex properties, resulting in the non-uniqueness or incomparability of the feature extraction results. The nonlinearity or non-convexity of features implies that the shape or structure of the feature space does not conform to the linear or convex assumptions, for example, curves, surfaces, or multi-peaks.

- The features of CMOPs often change with the problem, requiring the feature extraction method to have adaptability and generalization. The change of features indicates that the feature space changes with the input, output, parameters, objectives, and constraints of the problem, for example, dynamic, uncertain, or multimodal.

No universal feature extraction method exists that extracts the features of all constrained multi-objective optimization problems. To address this problem, we propose borrowing the word vector generation methods from natural language processing, for example, Word2Vec [159], FastText [160], ELMo [161], and BERT [162]. We believe that BERT is a promising direction, as it constructs a similar CMOPs feature extraction pre-training model that meets various downstream CMOPs tasks. We believe that graph neural network (GNN) [163] better represents the relationships between individuals in the population, and GNN is a promising direction. GNN is a kind of neural network that handles graph-structured data, using the topology and node attributes of the graph to learn the feature representation of the nodes. We view the population of CMOPs as a graph, in which each individual is a node, the similarity or distance between each individual is an edge, the objective value or constraint violation of each individual is a node attribute, and use GNN to learn the feature representation of the population, extracting features that are helpful for CMOPs solutions.

4.2. Autonomous recommendation of constrained multi-objective optimization algorithms

According to the no free lunch theorem [88], no algorithm has a performance advantage on one class of problems, that does not turn into a disadvantage on another class of problems. Therefore, to solve different constrained multi-objective optimization problems, it is necessary to select appropriate algorithms according to the characteristics of the problems. Currently, there are two types of methods for autonomous recommendation of algorithms, one is to automatically match the corresponding algorithms for different problems [76,77,79], the other is to automatically match the corresponding constraint handling techniques or meta-heuristic algorithms according to the state of the population during the evolution [19,78,80]. These two methods have their own advantages and disadvantages: the former can adapt to diverse problems, but requires a lot of prior knowledge and human intervention; the latter can adaptively adjust the algorithm parameters and strategies, but requires longer running time and more computing resources. However, there is no algorithm that combines these two methods, which may improve the efficiency and effectiveness of autonomous recommendation. With the development of deep learning, large language models have achieved remarkable results in the field of natural language processing. Liu et al. [89] combined large language models with multi-objective optimization, using suitable prompt engineering, to make the general large language models act as black-box search operators for MOEA/D with zero-shot learning, showing powerful performance. Their method can effectively handle the balance between constraints and objectives in multi-objective optimization problems, while using the strong expressive power of large language models to improve the search efficiency and quality of solutions. However, there is still a lack of research on the combination of constrained multi-objective optimization and large language models. We believe that this is a new direction, by combining large language models for algorithm autonomous recommendation. Specifically, we can consider the following aspects:

- How to use the pre-training and fine-tuning capabilities of large language models to extract the features and semantic information of constrained multi-objective optimization problems, to facilitate the decision-making of autonomous algorithm recommendation.
- How to design suitable prompts, so that the large language models can generate appropriate algorithms or parameters, as well as corresponding constraint handling techniques or meta-heuristic algorithms, according to the different problems and population states.
- How to evaluate and optimize the performance of the large language models on constrained multi-objective optimization problems.

4.3. Dynamic constrained multi-objective problems

Dynamic constrained multi-objective problems (DCMOPs) refer to multi-objective optimization problems in which the objective functions, constraints, or parameters change over time in dynamic environments. These problems have multiple conflicting objectives and constraints, and require the finding of optimal solutions under different environmental states. DCMOPs have a wide range of application scenarios in engineering, transportation, intelligent manufacturing, and so on, such as fluid catalytic cracking operation optimization [164], ore dressing process operation index optimization [165], and optimal control problems [166]. The main difficulty in solving DCMOPs is how to ensure the convergence and diversity of the population in dynamic environments. This means being able to track the dynamic Pareto optimal solution set in a timely manner, while maintaining the uniform and wide coverage of the solution set. At present, some dynamic CMOEAs have been proposed [167–170], that are mainly divided into two categories: diversity introduction methods and prediction methods [171]. Diversity introduction methods [172,173] improve the adaptability of the algorithm by increasing the diversity of the population, but may reduce the convergence speed. Prediction methods [174,175] use prediction models to predict the advantageous population in the changing environment, which can significantly improve the convergence of the population, but the performance of the prediction model is affected by the training cycle. Therefore, how to design more efficient algorithms is still a worthwhile research problem.

4.4. Dynamic interval constrained multi-objective problems

DCMOPs and interval multi-objective optimization problems (IMOPs) can be combined to form dynamic interval constrained multi-objective problems (DI-CMOPs). DI-CMOPs have applications in many practical scenarios, such as steelmaking continuous casting, hot rolling [176], robot path planning [177], multi-period portfolio selection problems [178,179]. The objective functions of these problems are usually interval values, which means that traditional methods, such as selection based on non-inferior solutions, detection and response of optimization problem changes, and so on, are not applicable to these problems. In addition, as the range of interval parameters changes over time, the non-inferior solutions of IMOPs may become inferior solutions in DI-CMOPs. Therefore, solving DI-CMOPs requires addressing the following challenges: accurately detecting the changes of optimization problems, quickly tracking the PF after the changes, and providing candidate solutions with good diversity and approximation in a timely manner [180].

4.5. Large-scale constrained multi-objective problems

Large-scale constrained multi-objective optimization problems (Large-Scale CMOPs) are optimization problems with a large number of decision variables and multiple conflicting objectives. These problems are very common in engineering, science, and management fields, such as multi-objective vehicle routing problem [181] and configuration software optimization [182], but traditional optimization methods or single evolutionary algorithms are often difficult to effectively solve them. Large-Scale CMOPs face the following main challenges:

- Exponential growth of search space: As the number of decision variables increases, the size of the search space grows exponentially, resulting in the sparsity, diversity reduction of feasible solutions, and the complexity, uncertainty increase of the PF.
- Difficulty of constraint handling: In Large-Scale CMOPs, the number and complexity of constraints may also be high, that leads to the uneven and discontinuous distribution of feasible solutions, and the fuzzy and irregular constraint boundaries. These make it difficult for evolutionary algorithms to satisfy and optimize the constraints. Moreover, the evaluation of constraints may also be very expensive, which limits the computational cost and efficiency of evolutionary algorithms.
- Interactivity among variables: In Large-Scale CMOPs, there may be strong correlation or dependency among decision variables, that affects the effectiveness of mutation and crossover operations of evolutionary algorithms, and the applicability of decomposition and dimensionality reduction techniques.
- Redundancy and irrelevance of high-dimensional features: In Large-Scale CMOPs, the dimension of decision variables is very high, but not all variables have significant impact on the objective functions and constraints, i.e., some variables are redundant or irrelevant. These variables increase the search difficulty and computational overhead of evolutionary algorithms, and reduce the performance and stability of evolutionary algorithms.

4.6. Robust constrained multi-objective problems

Robust constrained multi-objective problems (RC-MOPs) [183,184] are a type of optimization problem that aim to find a set of solutions that can satisfy multiple objective functions and constraints in uncertain environments. The goal of this kind of problem is to find solutions that are robust, i.e., that they can maintain the feasibility of the constraints and the quality of the objective functions even when the input data is noisy or perturbed. Solving this kind of problem faces the following challenges:

- How to define and measure robustness, i.e., how to quantify the degree of sensitivity of the solution to uncertainty, and how to assess the capability of the solution to withstand interference. Different definitions and measurement methods of robustness may result in different robust PFs.
- How to handle constraints, i.e., how to ensure the robustness and diversity of the solution while satisfying the constraints. The presence of constraints increases the complexity and discontinuity of the search space, thus increasing the difficulty of solving.

Therefore, it is necessary to design reasonable robustness mechanisms and robustness measurement methods, and combine them with constraint handling techniques, to balance convergence and diversity, and thus effectively solve robust CMOPs.

4.7. Computationally intensive constrained multi-objective problems

Computationally intensive constrained multi-objective problems (CICMOPs) are optimization problems with multiple possibly conflicting objective functions [185,186]. They require a large amount of CPU time or memory space to solve, and they must satisfy some constraints. The calculation process of the objective function or the constraint function is very complex, which makes this kind of problem difficult. It may involve high-dimensional numerical integration, nonlinear differential equation solving, accurate physical simulation, and other techniques. These calculation processes usually require a lot of iteration or recursion, resulting in exponential growth of computation time and space. A typical example is the computational fluid dynamics problem. It requires solving the Navier–Stokes equation, a nonlinear partial differential equation system that describes the motion of fluids [187]. To solve this kind of problem effectively, researchers have proposed specialized methods, such as the surrogate model-based method [63,188–191]. The idea of this method is to use simple functions to approximate the complex objective function or constraint function, thereby reducing the computation cost and improving the solution efficiency. However, since the surrogate model is only approximate, it may have errors, and it needs to be constantly updated and improved. The computation cost of CICMOPs is very high. Therefore, parallel execution is an effective way to speed up [134]. It can use multi-core processors or distributed systems to execute multiple computation tasks simultaneously, and it can improve its scalability. However, when studying the parallelization mechanism of surrogate models, we need to consider the knowledge transfer: how to share and transfer useful information between different computation nodes, such as samples or model parameters, to improve the accuracy and stability of surrogate models.

5. Conclusion

This paper provides a comprehensive review of constrained multi-objective optimization problems, involving the state-of-the-art algorithms, applications, and future research directions. Specifically, it proposes a classification method that can scientifically divide algorithms based on their characteristics, comprising three categories. This classification focuses on mathematical methods, evolutionary algorithms, and machine learning methods, analyzing the advantages and disadvantages of each CMOA. Then, it introduces and summarizes some practical applications of constrained multi-objective optimization, providing an application paradigm for extending CMOAs to real-world scenarios. Furthermore, potential future research directions are discussed. Overall, this work provides a systematic and scientific investigation of CMOPs, and helps researchers quickly preview the latest research status.

CRediT authorship contribution statement

Yuanyuan Hao: Supervision, Resources, Methodology, Writing – original draft, Writing – review & editing. **Chunliang Zhao:** Project administration, Methodology, Investigation. **Yiqin Zhang:** Data curation, Writing – review & editing. **Yuanze Cao:** Methodology, Investigation, Formal analysis, Data curation. **Zhong Li:** Supervision, Funding acquisition.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

Acknowledgments

This work was supported in part by the Shandong Provincial Natural Science Foundation under grant ZR2023QF065, the Natural Science Foundation of Fujian Province (No. 2020J01800), and the PPP Program of DAAD, Germany under the grant number 57653779.

References

[1] Q. Yu, C. Yang, G. Dai, L. Peng, X. Chen, Synchronous wireless sensor and sink placement method using dual-population coevolutionary constrained multi-objective optimization algorithm, IEEE Trans. Ind. Inform. (2022).

[2] S.Y.D. Sorkhabi, D.A. Romero, J.C. Beck, C.H. Amon, Constrained multi-objective wind farm layout optimization: Novel constraint handling approach based on constraint programming, Renew. Energy 126 (2018) 341–353.

[3] K. Lwin, R. Qu, G. Kendall, A learning-guided multi-objective evolutionary algorithm for constrained portfolio optimization, Appl. Soft Comput. 24 (2014) 757–772.

[4] Y. Wang, C. Zixing, Z. Yuren, X. Chixin, Constrained optimization evolutionary algorithms, J. Softw. 20 (1) (2009) 11–29.

[5] Z.-Z. Liu, Y. Wang, B.-C. Wang, Indicator-based constrained multiobjective evolutionary algorithms, IEEE Trans. Syst. Man. Cybern. Syst. 51 (9) (2021) 5414–5426, http://dx.doi.org/10.1109/TSMC.2019.2954491.

[6] W. Li, W. Gong, F. Ming, L. Wang, Constrained multi-objective evolutionary algorithm with an improved two-archive strategy, Knowl.-Based Syst. 246 (2022) 108732.

[7] M. Xia, M. Dong, A novel two-archive evolutionary algorithm for constrained multi-objective optimization with small feasible regions, Knowl.-Based Syst. 237 (2022) 107693.

[8] Z. Ma, Y. Wang, Evolutionary constrained multiobjective optimization: Test suite construction and performance comparisons, IEEE Trans. Evol. Comput. 23 (6) (2019) 972–986.

[9] Z. Fan, W. Li, X. Cai, H. Li, C. Wei, Q. Zhang, K. Deb, E. Goodman, Difficulty adjustable and scalable constrained multiobjective test problem toolkit, Evol. Comput. 28 (3) (2020) 339–378.

[10] Z. Fan, W. Li, X. Cai, H. Huang, Y. Fang, Y. You, J. Mo, C. Wei, E. Goodman, An improved epsilon constraint-handling method in MOEA/D for CMOPs with large infeasible regions, Soft Comput. 23 (2019) 12491–12510.

[11] Y.-T. Hsiao, H.-D. Chiang, C.-C. Liu, Y.-L. Chen, A computer package for optimal multi-objective VAR planning in large scale power systems, IEEE Trans. Power Syst. 9 (2) (1994) 668–676.

[12] L. Fu, J. Fan, L. Yuanchun, T. Yantao, D. Yisong, Study and application of a constrained multi-objective optimization algorithm, in: Proceedings of the IEEE International Vehicle Electronics Conference (IVEC'99)(Cat. No. 99EX257), IEEE, 1999, pp. 305–307.

[13] C.A.C. Coello, S.G. Brambila, J.F. Gamboa, M.G.C. Tapia, Multi-objective evolutionary algorithms: Past, present, and future, in: P.M. Pardalos, V. Rasskazova, M.N. Vrahatis (Eds.), Black Box Optimization, Machine Learning, and No-Free Lunch Theorems, Springer International Publishing, Cham, 2021, pp. 137–162.

[14] J. Liang, X. Ban, K. Yu, B. Qu, K. Qiao, C. Yue, K. Chen, K.C. Tan, A survey on evolutionary constrained multiobjective optimization, IEEE Trans. Evol. Comput. 27 (2) (2022) 201–221.

[15] G. Mavrotas, Effective implementation of the ε-constraint method in multi-objective mathematical programming problems, Appl. Math. Comput. 213 (2) (2009) 455–465, http://dx.doi.org/10.1016/j.amc.2009.03.037, URL: https://www.sciencedirect.com/science/article/pii/S0096300309002574.

[16] G. Yang, X. Li, L. Huo, Q. Liu, A solving approach for fuzzy multi-objective linear fractional programming and application to an agricultural planting structure optimization problem, Chaos Solitons Fractals 141 (2020) 110352.

[17] Y. Tian, T. Zhang, J. Xiao, X. Zhang, Y. Jin, A coevolutionary framework for constrained multiobjective optimization problems, IEEE Trans. Evol. Comput. 25 (1) (2020) 102–116.

[18] Z. Fan, W. Li, X. Cai, H. Li, C. Wei, Q. Zhang, K. Deb, E. Goodman, Push and pull search for solving constrained multi-objective optimization problems, Swarm Evol. Comput. 44 (2019) 665–679.

[19] M. Zuo, D. Gong, Y. Wang, X. Ye, B. Zeng, F. Meng, Process knowledge-guided autonomous evolutionary optimization for constrained multiobjective problems, IEEE Trans. Evol. Comput. (2023).

[20] F. Ming, W. Gong, S. Li, L. Wang, Z. Liao, Handling constrained many-objective optimization problems via determinantal point processes, Inform. Sci. (2023) 119260.

[21] G.B. Dantzig, Linear programming, Oper. Res. 50 (1) (2002) 42–47.

[22] M. Ehrgott, S. Ruzika, Improved ε-constraint method for multiobjective programming, J. Optim. Theory Appl. 138 (3) (2008) 375–396.

[23] W.A. Lodwick, J. Kacprzyk, Fuzzy Optimization: Recent Advances and Applications, vol. 254, Springer, 2010.

[24] B. Liu, F.V. Fernandez, P. Gao, G. Gielen, A fuzzy selection based constraint handling method for multi-objective optimization of analog cells, in: 2009 European Conference on Circuit Theory and Design, IEEE, 2009, pp. 611–614.

[25] C. Saha, S. Das, K. Pal, S. Mukherjee, A fuzzy rule-based penalty function approach for constrained evolutionary optimization, IEEE Trans. Cybern. 46 (12) (2014) 2953–2965.

[26] Z. Li, S. Chen, S. Zhang, S. Jiang, Y. Gu, M. Nouioua, FSB-EA: Fuzzy search bias guided constraint handling technique for evolutionary algorithm, Expert Syst. Appl. 119 (2019) 20–35.

[27] G. Cocchi, M. Lapucci, An augmented Lagrangian algorithm for multi-objective optimization, Comput. Optim. Appl. 77 (1) (2020) 29–56.

[28] M.A.T. Ansary, G. Panda, A sequential quadratic programming method for constrained multi-objective optimization problems, J. Appl. Math. Comput. 64 (1) (2020) 379–397.

[29] X. Yang, Z. Leng, S. Xu, C. Yang, L. Yang, K. Liu, Y. Song, L. Zhang, Multi-objective optimal scheduling for CCHP microgrids considering peak-load reduction by augmented ε-constraint method, Renew. Energy 172 (2021) 408–423.

[30] D. Han, W. Du, Y. Jin, W. Du, G. Yu, A fuzzy constraint handling technique for decomposition-based constrained multi-and many-objective optimization, Inf. Sci. 597 (2022) 318–340.

[31] K.K. Lai, J.K. Maurya, S.K. Mishra, Multiobjective approximate gradient projection method for constrained vector optimization: Sequential optimality conditions without constraint qualifications, J. Comput. Appl. Math. 410 (2022) 114122.

[32] B. Pérez-Cañedo, J.L. Verdegay, E.R. Concepción-Morales, et al., An ε-constraint method for multiobjective linear programming in intuitionistic fuzzy environment, Int. J. Intell. Syst. 2023 (2023).

[33] K. Sharma, V.P. Singh, A. Ebrahimnejad, D. Chakraborty, Solving a multi-objective chance constrained hierarchical optimization problem under intuitionistic fuzzy environment with its application, Expert Syst. Appl. 217 (2023) 119595.

[34] B. Alouane, M. Boulif, Fuzzy constraint prioritization to solve heavily constrained problems with the genetic algorithm, Eng. Appl. Artif. Intell. 119 (2023) 105768.

[35] M. Ming, A. Trivedi, R. Wang, D. Srinivasan, T. Zhang, A dual-population-based evolutionary algorithm for constrained multiobjective optimization, IEEE Trans. Evol. Comput. 25 (4) (2021) 739–753.

[36] Z.-Z. Liu, B.-C. Wang, K. Tang, Handling constrained multiobjective optimization problems via bidirectional coevolution, IEEE Trans. Cybern. 52 (10) (2021) 10163–10176.

[37] R. Jiao, B. Xue, M. Zhang, A multiform optimization framework for constrained multiobjective optimization, IEEE Trans. Cybern. (2022).

[38] K. Qiao, J. Liang, K. Yu, M. Wang, B. Qu, C. Yue, Y. Guo, A self-adaptive evolutionary multi-task based constrained multi-objective evolutionary algorithm, IEEE Trans. Emerg. Top. Computat. Intell. (2023).

[39] L. Li, C. He, J. Lin, L. Pan, Two-archive-based competitive swarm optimizer for constrained multiobjective optimization, in: 2023 5th International Conference on Data-Driven Optimization of Complex Systems, DOCS, 2023, pp. 1–6, http://dx.doi.org/10.1109/DOCS60977.2023.10294727.

[40] Y. Yang, J. Liu, S. Tan, A multi-objective evolutionary algorithm for steady-state constrained multi-objective optimization problems, Appl. Soft Comput. 101 (2021) 107042.

[41] M. Li, S. Yang, X. Liu, Pareto or non-Pareto: Bi-criterion evolution in multiobjective optimization, IEEE Trans. Evol. Comput. 20 (5) (2015) 645–665.

[42] J. Wang, Y. Li, Q. Zhang, Z. Zhang, S. Gao, Cooperative multiobjective evolutionary algorithm with propulsive population for constrained multiobjective optimization, IEEE Trans. Syst. Man. Cybern. Syst. 52 (6) (2021) 3476–3491.

[43] K. Qiao, K. Yu, B. Qu, J. Liang, H. Song, C. Yue, An evolutionary multitasking optimization framework for constrained multiobjective optimization problems, IEEE Trans. Evol. Comput. 26 (2) (2022) 263–277.

[44] F. Ming, W. Gong, L. Wang, C. Lu, A tri-population based co-evolutionary framework for constrained multi-objective optimization problems, Swarm Evol. Comput. 70 (2022) 101055.

[45] Y. Li, X. Feng, H. Yu, A constrained multiobjective evolutionary algorithm with the two-archive weak cooperation, Inform. Sci. 615 (2022) 415–430.

[46] X. Kong, Y. Yang, Z. Lv, J. Zhao, R. Fu, A dynamic dual-population co-evolution multi-objective evolutionary algorithm for constrained multi-objective optimization problems, Appl. Soft Comput. 141 (2023) 110311.

[47] J. Cao, Z. Yan, Z. Chen, J. Zhang, A coevolutionary constrained multi-objective algorithm with a learning constraint boundary, Appl. Soft Comput. 148 (2023) 110845, http://dx.doi.org/10.1016/j.asoc.2023.110845, URL: https://www.sciencedirect.com/science/article/pii/S1568494623008633.

[48] S. Song, K. Zhang, L. Zhang, N. Wu, A dual-population algorithm based on self-adaptive epsilon method for constrained multi-objective optimization, Inform. Sci. 655 (2024) 119906.

[49] R. Jiao, S. Zeng, C. Li, A feasible-ratio control technique for constrained optimization, Inform. Sci. 502 (2019) 201–217.

[50] Z. Fan, Z. Wang, W. Li, Y. Yuan, Y. You, Z. Yang, F. Sun, J. Ruan, Push and pull search embedded in an M2M framework for solving constrained multi-objective optimization problems, Swarm Evol. Comput. 54 (2020) 100651.

[51] K. Yu, J. Liang, B. Qu, C. Yue, Purpose-directed two-phase multiobjective differential evolution for constrained multiobjective optimization, Swarm Evol. Comput. 60 (2021) 100799.

[52] Y. Tian, Y. Zhang, Y. Su, X. Zhang, K.C. Tan, Y. Jin, Balancing objective optimization and constraint satisfaction in constrained evolutionary multiobjective optimization, IEEE Trans. Cybern. 52 (9) (2021) 9559–9572.

[53] K. Yu, J. Liang, B. Qu, Y. Luo, C. Yue, Dynamic selection preference-assisted constrained multiobjective differential evolution, IEEE Trans. Syst. Man. Cybern. Syst. 52 (5) (2021) 2954–2965.

[54] F. Ming, W. Gong, H. Zhen, S. Li, L. Wang, Z. Liao, A simple two-stage evolutionary algorithm for constrained multi-objective optimization, Knowl.-Based Syst. 228 (2021) 107263.
[55] H. Xu, X. Cai, Z. Li, Z. Fan, A two-phase constrained multi-objective evolutionary algorithm based on the constrained decomposition approach, in: 2021 IEEE 7th International Conference on Cloud Computing and Intelligent Systems, CCIS, IEEE, 2021, pp. 154–158.
[56] H. Ma, H. Wei, Y. Tian, R. Cheng, X. Zhang, A multi-stage evolutionary algorithm for multi-objective optimization with complex constraints, Inform. Sci. 560 (2021) 68–91.
[57] R. Jiao, S. Zeng, C. Li, Y.-S. Ong, Two-type weight adjustments in MOEA/D for highly constrained many-objective optimization, Inform. Sci. 578 (2021) 592–614.
[58] X. Feng, A. Pan, Z. Ren, Z. Fan, Hybrid driven strategy for constrained evolutionary multi-objective optimization, Inform. Sci. 585 (2022) 344–365.
[59] K. Zhang, Z. Xu, G.G. Yen, L. Zhang, Two-stage multi-objective evolution strategy for constrained multi-objective optimization, IEEE Trans. Evol. Comput. (2022).
[60] R. Sun, J. Zou, Y. Liu, S. Yang, J. Zheng, A multi-stage algorithm for solving multi-objective optimization problems with multi-constraints, IEEE Trans. Evol. Comput. (2022).
[61] E. Song, G. Han, G. Zhang, A two-stage differential evolution for constrained multi-objective optimization problems, in: 2022 18th International Conference on Computational Intelligence and Security, CIS, IEEE, 2022, pp. 12–17.
[62] J. Liang, K. Qiao, K. Yu, B. Qu, C. Yue, W. Guo, L. Wang, Utilizing the relationship between unconstrained and constrained pareto fronts for constrained multiobjective optimization, IEEE Trans. Cybern. (2022).
[63] Y. Liu, J. Liu, Y. Jin, F. Li, T. Zheng, A surrogate-assisted two-stage differential evolution for expensive constrained optimization, IEEE Trans. Emerg. Top. Computat. Intell. (2023).
[64] J. Zou, J. Luo, Y. Liu, S. Yang, J. Zheng, A flexible two-stage constrained multi-objective evolutionary algorithm based on automatic regulation, Inform. Sci. 634 (2023) 227–243.
[65] M. Xia, Q. Chong, M. Dong, A constrained multi-objective evolutionary algorithm with two-stage resources allocation, Swarm Evol. Comput. 79 (2023) 101313.
[66] J. Feng, S. Liu, S. Yang, J. Zheng, J. Liu, An adaptive tradeoff evolutionary algorithm with composite differential evolution for constrained multi-objective optimization, Swarm Evol. Comput. 83 (2023) 101386.
[67] G. Chen, Y. Guo, Y. Wang, J. Liang, D. Gong, S. Yang, Evolutionary dynamic constrained multiobjective optimization: Test suite and algorithm, IEEE Trans. Evol. Comput. (2023).
[68] W. Huang, J. Zou, Y. Liu, S. Yang, J. Zheng, Global and local feasible solution search for solving constrained multi-objective optimization, Inform. Sci. 649 (2023) 119467.
[69] D. Yang, G. Li, G. Cheng, On the efficiency of chaos optimization algorithms for global optimization, Chaos Solitons Fractals 34 (4) (2007) 1366–1375.
[70] N. Aslimani, T. El-Ghazali, R. Ellaia, A new chaotic-based approach for multi-objective optimization, Algorithms 13 (9) (2020) 204.
[71] Q. Li, L. Liu, X. Yuan, et al., Multiobjective parallel chaos optimization algorithm with crossover and merging operation, Math. Probl. Eng. 2020 (2020).
[72] L. Zhang, R. Yu, J. Liu, Y. Xu, H. Jiang, P. Xu, P. Lin, Y. Luo, A constrained multi-objective evolutionary algorithm based on weak cooperation framework and multi-chaotic operators, in: 2022 Global Conference on Robotics, Artificial Intelligence and Information Technology, GCRAIT, 2022, pp. 102–106, http://dx.doi.org/10.1109/GCRAIT55928.2022.00030.
[73] Y. Takubo, M. Kanazaki, Robust constrained multi-objective evolutionary algorithm based on polynomial chaos expansion for trajectory optimization, in: 2022 IEEE Congress on Evol. Comput., CEC, IEEE, 2022, pp. 1–9.
[74] S. Yacoubi, G. Manita, A. Chhabra, O. Korbaa, S. Mirjalili, A multi-objective chaos game optimization algorithm based on decomposition and random learning mechanisms for numerical optimization, Appl. Soft Comput. 144 (2023) 110525.
[75] H.T. Rauf, J. Gao, A. Almadhor, A. Haider, Y.-D. Zhang, F. Al-Turjman, Multi population-based chaotic differential evolution for multi-modal and multi-objective optimization problems, Appl. Soft Comput. 132 (2023) 109909.
[76] Y. Tian, S. Peng, T. Rodemann, X. Zhang, Y. Jin, Automated selection of evolutionary multi-objective optimization algorithms, in: 2019 IEEE Symposium Series on Computational Intelligence, SSCI, IEEE, 2019, pp. 3225–3232.
[77] K. Qiao, K. Yu, B. Qu, J. Liang, C. Yue, X. Ban, Feature extraction for recommendation of constrained multi-objective evolutionary algorithms, IEEE Trans. Evol. Comput. (2022).
[78] Z. Han, Y. Yang, H. Ye, A deep reinforcement learning based multiple meta-heuristic methods approach for resource constrained multi-project scheduling problem, in: 2022 7th International Conference on Intelligent Computing and Signal Processing, ICSP, IEEE, 2022, pp. 26–29.
[79] Y. Wang, M. Zuo, D. Gong, Migration-based algorithm library enrichment for constrained multi-objective optimization and applications in algorithm selection, Inform. Sci. 649 (2023) 119593.
[80] F. Ming, W. Gong, L. Gao, Adaptive auxiliary task selection for multitasking-assisted constrained multi-objective optimization [feature], IEEE Comput. Intell. Mag. 18 (2) (2023) 18–30.
[81] A. Fallahi, E.A. Bani, S.T.A. Niaki, A constrained multi-item EOQ inventory model for reusable items: Reinforcement learning-based differential evolution and particle swarm optimization, Expert Syst. Appl. 207 (2022) 118018.
[82] X. Yu, P. Xu, F. Wang, X. Wang, Reinforcement learning-based differential evolution algorithm for constrained multi-objective optimization problems, Eng. Appl. Artif. Intell. 131 (2024) 107817.
[83] L. Ma, N. Li, Y. Guo, X. Wang, S. Yang, M. Huang, H. Zhang, Learning to optimize: reference vector reinforcement learning adaption to constrained many-objective optimization of industrial copper burdening system, IEEE Trans. Cybern. (2021).
[84] C. Wang, H. Huang, X. Zhang, Growing neural gas network for offspring generation in evolutionary constrained multi-objective optimization, IEEE Trans. Emerg. Top. Computat. Intell. (2023).
[85] A. Kulesza, B. Taskar, et al., Determinantal point processes for machine learning, Found. Trends Mach. Learn. 5 (2–3) (2012) 123–286.
[86] Z. Hu, W. Gong, W. Pedrycz, Y. Li, Deep reinforcement learning assisted co-evolutionary differential evolution for constrained optimization, Swarm Evol. Comput. 83 (2023) 101387.
[87] Z. Liu, P. Zhao, J. Cao, J. Zhang, Z. Chen, A constrained multi-objective evolutionary algorithm with Pareto estimation via neural network, Expert Syst. Appl. 237 (2024) 121718.
[88] D.H. Wolpert, W.G. Macready, No free lunch theorems for optimization, IEEE Trans. Evol. Comput. 1 (1) (1997) 67–82.
[89] F. Liu, X. Lin, Z. Wang, S. Yao, X. Tong, M. Yuan, Q. Zhang, Large language model for multi-objective evolutionary optimization, 2023, arXiv preprint arXiv:2310.12541.
[90] T. Ray, K. Tai, K.C. Seow, Multiobjective design optimization by an evolutionary algorithm, Eng. Optim. 33 (4) (2001) 399–424.
[91] Q. Bao, M. Wang, G. Dai, X. Chen, Z. Song, S. Li, An archive-based two-stage evolutionary algorithm for constrained multi-objective optimization problems, Swarm Evol. Comput. 75 (2022) 101161.
[92] G.R. Kocis, I.E. Grossmann, Global optimization of nonconvex mixed-integer nonlinear programming (MINLP) problems in process synthesis, Ind. Eng. Chem. Res. 27 (8) (1988) 1407–1421.
[93] Y. Xiang, X. Yang, Y. Zhou, H. Huang, Enhancing decomposition-based algorithms by estimation of distribution for constrained optimal software product selection, IEEE Trans. Evol. Comput. 24 (2) (2019) 245–259.
[94] Y. Xiang, X. Yang, H. Huang, J. Wang, Balancing constraints and objectives by considering problem types in constrained multiobjective optimization, IEEE Trans. Cybern. (2021).
[95] L.V. Santana-Quintero, A.G. Hernández-Díaz, J. Molina, C.A.C. Coello, R. Caballero, DEMORS: A hybrid multi-objective optimization algorithm using differential evolution and rough set theory for constrained problems, Comput. Oper. Res. 37 (3) (2010) 470–480.
[96] C. Qin, F. Ming, W. Gong, Q. Gu, Constrained multi-objective optimization via two archives assisted push–pull evolutionary algorithm, Swarm Evol. Comput. 75 (2022) 101178.
[97] M. Kohli, S. Arora, Chaotic grey wolf optimization algorithm for constrained optimization problems, J. Comput. Des. Eng. 5 (4) (2018) 458–472.
[98] L. Vallozzi, W. Vandendriessche, H. Rogier, C. Hertleer, M. Scarpello, Design of a protective garment GPS antenna, Microw. Opt. Technol. Lett. 51 (6) (2009) 1504–1508.
[99] R. Saravanan, S. Ramabalan, N.G.R. Ebenezer, C. Dharmaraja, Evolutionary multi criteria design optimization of robot grippers, Appl. Soft Comput. 9 (1) (2009) 159–172.
[100] Y. Yang, J. Liu, S. Tan, H. Wang, Application of constrained multi-objective evolutionary algorithm in multi-source compressed-air pipeline optimization problems, IFAC-PapersOnLine 51 (18) (2018) 168–173.
[101] Y. Yang, J. Liu, S. Tan, Y. Liu, A multi-objective differential evolution algorithm based on domination and constraint-handling switching, Inform. Sci. 579 (2021) 796–813.
[102] Y. Yang, J. Liu, S. Tan, A partition-based constrained multi-objective evolutionary algorithm, Swarm Evol. Comput. 66 (2021) 100940.
[103] Q. Zhang, Q. Wang, G. Li, Switched system identification based on the constrained multi-objective optimization problem with application to the servo turntable, Int. J. Control Autom. Syst. 14 (5) (2016) 1153–1159.
[104] H. Mala-Jetmarova, N. Sultanova, D. Savic, Lost in optimisation of water distribution systems? A literature review of system operation, Environ. Model. Softw. 93 (2017) 209–254.
[105] Q. Ouyang, W. Lu, Z. Hou, Y. Zhang, S. Li, J. Luo, Chance-constrained multi-objective optimization of groundwater remediation design at DNAPLs-contaminated sites using a multi-algorithm genetically adaptive method, J. Contam. Hydrol. 200 (2017) 15–23.
[106] J. Marti, L. Geissbühler, V. Becattini, A. Haselbacher, A. Steinfeld, Constrained multi-objective optimization of thermocline packed-bed thermal-energy storage, Appl. Energy 216 (2018) 694–708.

[107] Z. Su, G. Zhang, F. Yue, D. Zhan, M. Li, B. Li, X. Yao, Enhanced constraint handling for reliability-constrained multiobjective testing resource allocation, IEEE Trans. Evol. Comput. 25 (3) (2021) 537–551.
[108] G. Dhiman, V. Kumar, Multi-objective spotted hyena optimizer: a multi-objective optimization algorithm for engineering problems, Knowl.-Based Syst. 150 (2018) 175–197.
[109] F. Ming, W. Gong, L. Wang, L. Gao, A constrained many-objective optimization evolutionary algorithm with enhanced mating and environmental selections, IEEE Trans. Cybern. (2022).
[110] J. Shi, M. Chen, Y. Ma, F. Qiao, A new boredom-aware dual-resource constrained flexible job shop scheduling problem using a two-stage multi-objective particle swarm optimization algorithm, Inform. Sci. 643 (2023) 119141.
[111] Y. Hou, Y. Zhang, N. Wu, Q. Zhu, Constrained multi-objective optimization of short-term crude oil scheduling with dual pipelines and charging tank maintenance requirement, Inform. Sci. 588 (2022) 381–404.
[112] X. Zuo, C. Chen, W. Tan, M. Zhou, Vehicle scheduling of an urban bus line via an improved multiobjective genetic algorithm, IEEE Trans. Intell. Transp. Syst. 16 (2) (2014) 1030–1041.
[113] Z. Ma, Y. Wang, Shift-based penalty for evolutionary constrained multiobjective optimization and its application, IEEE Trans. Cybern. (2021).
[114] U. Güvenc, Y. Sönmez, S. Duman, N. Yörükeren, Combined economic and emission dispatch solution using gravitational search algorithm, Sci. Iran. 19 (6) (2012) 1754–1762.
[115] M.-R. Chen, G.-Q. Zeng, K.-D. Lu, Constrained multi-objective population extremal optimization based economic-emission dispatch incorporating renewable energy resources, Renew. Energy 143 (2019) 277–294.
[116] M.A. El-Shorbagy, A. Mousa, Constrained multiobjective equilibrium optimizer algorithm for solving combined economic emission dispatch problem, Complexity 2021 (2021) 1–14.
[117] X. Yu, X. Yu, Y. Lu, G.G. Yen, M. Cai, Differential evolution mutation operators for constrained multi-objective optimization, Appl. Soft Comput. 67 (2018) 452–466.
[118] S.M. Mohseni-Bonab, A. Rabiee, B. Mohammadi-Ivatloo, Voltage stability constrained multi-objective optimal reactive power dispatch under load and wind power uncertainties: A stochastic approach, Renew. Energy 85 (2016) 598–609.
[119] C. Peng, S. Qiu, A decomposition-based constrained multi-objective evolutionary algorithm with a local infeasibility utilization mechanism for UAV path planning, Appl. Soft Comput. 118 (2022) 108495.
[120] X. Wang, J. Zhao, C. Cheng, M. Qi, A multi-objective fuzzy facility location problem with congestion and priority for drone-based emergency deliveries, Comput. Ind. Eng. 179 (2023) 109167.
[121] R. Latpate, S.S. Kurade, Multi-objective multi-index transportation model for crude oil using fuzzy NSGA-II, IEEE Trans. Intell. Transp. Syst. 23 (2) (2020) 1347–1356.
[122] C.-T. Chen, H.-V. Pham, Trajectory planning in parallel kinematic manipulators using a constrained multi-objective evolutionary algorithm, Nonlinear Dynam. 67 (2012) 1669–1681.
[123] S. Maity, A. Roy, M. Maiti, A rough multi-objective genetic algorithm for uncertain constrained multi-objective solid travelling salesman problem, Granul. Comput. 4 (1) (2019) 125–142.
[124] G. Kocis, I. Grossmann, A modelling and decomposition strategy for the minlp optimization of process flowsheets, Comput. Chem. Eng. 13 (7) (1989) 797–819, http://dx.doi.org/10.1016/0098-1354(89)85053-7, URL: https://www.sciencedirect.com/science/article/pii/0098135489850537.
[125] J. Zou, R. Sun, S. Yang, J. Zheng, A dual-population algorithm based on alternative evolution and degeneration for solving constrained multi-objective optimization problems, Inform. Sci. 579 (2021) 89–102.
[126] J. Zou, R. Sun, Y. Liu, Y. Hu, S. Yang, J. Zheng, K. Li, A multi-population evolutionary algorithm using new cooperative mechanism for solving multi-objective problems with multi-constraint, IEEE Trans. Evol. Comput. (2023).
[127] X. Zhu, C. Jiang, K. Hao, R. Wang, High-dimensional interactive adaptive RVEA for multi-objective optimization of polyester polymerization process, Inform. Sci. 650 (2023) 119707.
[128] A. Kumar, G. Wu, M.Z. Ali, Q. Luo, R. Mallipeddi, P.N. Suganthan, S. Das, A benchmark-suite of real-world constrained multi-objective optimization problems and some baseline results, Swarm Evol. Comput. 67 (2021) 100961.
[129] A. Kumar, G. Wu, M.Z. Ali, R. Mallipeddi, P.N. Suganthan, S. Das, A test-suite of non-convex constrained optimization problems from the real-world and some baseline results, Swarm Evol. Comput. 56 (2020) 100693.
[130] M.A. Tawhid, V. Savsani, A novel multi-objective optimization algorithm based on artificial algae for multi-objective engineering design problems, Appl. Intell. 48 (2018) 3762–3781.
[131] M. Kotinis, A particle swarm optimizer for constrained multi-objective engineering design problems, Eng. Optim. 42 (10) (2010) 907–926.
[132] G. Chiandussi, M. Codegone, S. Ferrero, F.E. Varesio, Comparison of multi-objective optimization methodologies for engineering applications, Comput. Math. Appl. 63 (5) (2012) 912–942.
[133] H. Jain, K. Deb, An evolutionary many-objective optimization algorithm using reference-point based nondominated sorting approach, part II: Handling constraints and extending to an adaptive approach, IEEE Trans. Evol. Comput. 18 (4) (2013) 602–622.
[134] C. He, Y. Zhang, D. Gong, X. Ji, A review of surrogate-assisted evolutionary algorithms for expensive optimization problems, Expert Syst. Appl. (2023) 119495.
[135] H.-Z. Huang, Y.-K. Gu, X. Du, An interactive fuzzy multi-objective optimization method for engineering design, Eng. Appl. Artif. Intell. 19 (5) (2006) 451–460.
[136] M.G. Parsons, R.L. Scott, Formulation of multicriterion design optimization problems for solution with scalar numerical optimization methods, J. Ship Res. 48 (01) (2004) 61–76.
[137] B. Kannan, S.N. Kramer, An augmented Lagrange multiplier based method for mixed integer discrete continuous optimization and its applications to mechanical design, 1994.
[138] A.M. Gujarathi, S. Purohit, B. Srikanth, Optimization of reactor network design problem using jumping gene adaptation of differential evolution, in: Journal of Physics: Conference Series, vol. 622, no. 1, IOP Publishing, 2015, 012044.
[139] A. Gadagi, C. Adake, A constrained multi-objective optimization of turning process parameters by genetic algorithm and particle swarm optimization techniques, Mater. Today: Proc. 42 (2021) 1207–1212.
[140] H. Yue, C. Guo, Q. Li, L. Zhao, G. Hao, Milling parameters optimization of Al-Li alloy thin-wall workpieces using response surface methodology and particle swarm optimization, CMES Comput. Model. Eng. Sci. 124 (3) (2020) 937–952.
[141] Y. Yang, P.-Q. Huang, X. Kong, J. Zhao, A constrained multi-objective evolutionary algorithm assisted by an additional objective function, Appl. Soft Comput. 132 (2023) 109904.
[142] Q. Gu, J. Bai, X. Li, N. Xiong, C. Lu, A constrained multi-objective evolutionary algorithm based on decomposition with improved constrained dominance principle, Swarm Evol. Comput. 75 (2022) 101162.
[143] J. Yuan, H.-L. Liu, Z. He, A constrained multi-objective evolutionary algorithm using valuable infeasible solutions, Swarm Evol. Comput. 68 (2022) 101020.
[144] K. Musselman, J. Talavage, A tradeoff cut approach to multiple objective optimization, Oper. Res. 28 (6) (1980) 1424–1435.
[145] E.M. Golafshani, M. Arashpour, A. Kashani, Green mix design of rubbercrete using machine learning-based ensemble model and constrained multi-objective optimization, J. Clean. Prod. 327 (2021) 129518.
[146] Y. Wang, H. Hu, X. Sun, Y. Zhang, D. Gong, Unified operation optimization model of integrated coal mine energy systems and its solutions based on autonomous intelligence, Appl. Energy 328 (2022) 120106.
[147] J.-F. Qiao, Y. Hou, L. Zhang, H.-G. Han, Adaptive fuzzy neural network control of wastewater treatment process with multiobjective operation, Neurocomputing 275 (2018) 383–393.
[148] M. Ming, R. Wang, H. Ishibuchi, T. Zhang, A novel dual-stage dual-population evolutionary algorithm for constrained multiobjective optimization, IEEE Trans. Evol. Comput. 26 (5) (2021) 1129–1143.
[149] T. Ray, K. Tai, C. Seow, An evolutionary algorithm for multiobjective optimization, Eng. Optim. 33 (3) (2001) 399–424.
[150] Y. Hou, Y. Wu, H. Han, Multistate-constrained multiobjective differential evolution algorithm with variable neighborhood strategy, IEEE Trans. Cybern. (2022).
[151] X. Yin, Z. Zhao, W. Yang, Optimizing cleaner productions of sustainable energies: A co-design framework for complementary operations of offshore wind and pumped hydro-storages, J. Clean. Prod. 396 (2023) 135832.
[152] L.H. Lee, K.C. Tan, K. Ou, Y.H. Chew, Vehicle capacity planning system: A case study on vehicle routing problem with time windows, IEEE Trans. Syst. Man Cybern. A 33 (2) (2003) 169–178.
[153] A.K. Rathore, J. Holtz, T. Boller, Synchronous optimal pulsewidth modulation for low-switching-frequency control of medium-voltage multilevel inverters, IEEE Trans. Ind. Electron. 57 (7) (2010) 2374–2381.
[154] A.K. Rathore, J. Holtz, T. Boller, Optimal pulsewidth modulation of multilevel inverters for low switching frequency control of medium voltage high power industrial AC drives, in: 2010 IEEE Energy Conversion Congress and Exposition, IEEE, 2010, pp. 4569–4574.
[155] A. Edpuganti, A.K. Rathore, Optimal pulsewidth modulation for common-mode voltage elimination scheme of medium-voltage modular multilevel converter-fed open-end stator winding induction motor drives, IEEE Trans. Ind. Electron. 64 (1) (2016) 848–856.
[156] G. Yang, X. Li, L. Huo, Q. Liu, A solving approach for fuzzy multi-objective linear fractional programming and application to an agricultural planting structure optimization problem, Chaos Solitons Fractals 141 (2020) 110352.
[157] D. Ozgen, B. Gulsun, Combining possibilistic linear programming and fuzzy AHP for solving the multi-objective capacitated multi-facility location problem, Inform. Sci. 268 (2014) 185–201.
[158] P. Peykani, M. Nouri, F. Eshghi, M. Khamechian, H. Farrokhi-Asl, A novel mathematical approach for fuzzy multi-period multi-objective portfolio optimization problem under uncertain environment and practical constraints, J. Fuzzy Ext. Appl. 2 (3) (2021) 191–203.
[159] T. Mikolov, K. Chen, G. Corrado, J. Dean, Efficient estimation of word representations in vector space, 2013, arXiv preprint arXiv:1301.3781.
[160] A. Joulin, E. Grave, P. Bojanowski, T. Mikolov, Bag of tricks for efficient text classification, 2016, arXiv preprint arXiv:1607.01759.
[161] S. Ilić, E. Marrese-Taylor, J.A. Balazs, Y. Matsuo, Deep contextualized word representations for detecting sarcasm and irony, 2018, arXiv preprint arXiv:1809.09795.

[162] J. Devlin, M.-W. Chang, K. Lee, K. Toutanova, Bert: Pre-training of deep bidirectional transformers for language understanding, 2018, arXiv preprint arXiv:1810.04805.
[163] F. Scarselli, M. Gori, A.C. Tsoi, M. Hagenbuchner, G. Monfardini, The graph neural network model, IEEE Trans. Neural Netw. 20 (1) (2008) 61–80.
[164] Q. Chen, J. Ding, T. Chai, Q. Pan, Evolutionary optimization under uncertainty: The strategies to handle varied constraints for fluid catalytic cracking operation, IEEE Trans. Cybern. 52 (4) (2020) 2249–2262.
[165] C. Yang, J. Ding, Constrained dynamic multi-objective evolutionary optimization for operational indices of beneficiation process, J. Intell. Manuf. 30 (2019) 2701–2713.
[166] M. Schlegel, W. Marquardt, Adaptive switching structure detection for the solution of dynamic optimization problems, Ind. Eng. Chem. Res. 45 (24) (2006) 8083–8094.
[167] X. Cai, C. Xia, Q. Zhang, Z. Mei, H. Hu, L. Wang, J. Hu, The collaborative local search based on dynamic-constrained decomposition with grids for combinatorial multiobjective optimization, IEEE Trans. Cybern. 51 (5) (2019) 2639–2650.
[168] K. Qiao, K. Yu, B. Qu, J. Liang, H. Song, C. Yue, H. Lin, K.C. Tan, Dynamic auxiliary task-based evolutionary multitasking for constrained multi-objective optimization, IEEE Trans. Evol. Comput. (2022).
[169] R. Azzouz, S. Bechikh, L.B. Said, W. Trabelsi, Handling time-varying constraints and objectives in dynamic evolutionary multi-objective optimization, Swarm Evol. Comput. 39 (2018) 222–248.
[170] Q. Chen, J. Ding, S. Yang, T. Chai, A novel evolutionary algorithm for dynamic constrained multiobjective optimization problems, IEEE Trans. Evol. Comput. 24 (4) (2019) 792–806.
[171] Z. Liang, T. Wu, X. Ma, Z. Zhu, S. Yang, A dynamic multiobjective evolutionary algorithm based on decision variable classification, IEEE Trans. Cybern. 52 (3) (2020) 1602–1615.
[172] K. Deb, U.B. Rao N, S. Karthik, Dynamic multi-objective optimization and decision-making using modified NSGA-II: a case study on hydro-thermal power scheduling, in: International Conference on Evolutionary Multi-Criterion Optimization, Springer, 2007, pp. 803–817.
[173] C.-K. Goh, K.C. Tan, A competitive-cooperative coevolutionary paradigm for dynamic multiobjective optimization, IEEE Trans. Evol. Comput. 13 (1) (2008) 103–127.
[174] A. Zhou, Y. Jin, Q. Zhang, A population prediction strategy for evolutionary dynamic multiobjective optimization, IEEE Trans. Cybern. 44 (1) (2013) 40–53.
[175] A. Muruganantham, K.C. Tan, P. Vadakkepat, Evolutionary dynamic multiobjective optimization via kalman filter prediction, IEEE Trans. Cybern. 46 (12) (2015) 2862–2873.
[176] J. Lin, M. Liu, J. Hao, S. Jiang, A multi-objective optimization approach for integrated production planning under interval uncertainties in the steel industry, Comput. Oper. Res. 72 (2016) 189–203.
[177] Y. Zhang, D.-w. Gong, J.-h. Zhang, Robot path planning in uncertain environment using multi-objective particle swarm optimization, Neurocomputing 103 (2013) 172–185.
[178] Y.-J. Liu, W.-G. Zhang, J.-B. Wang, Multi-period cardinality constrained portfolio selection models with interval coefficients, Ann. Oper. Res. 244 (2016) 545–569.
[179] J. Sun, X. Yan, Z. Heng, L. Zhiping, Interval multi-objective programming methods for solving multi-period portfolio selection problems, Control Decis. 35 (3) (2020) 645–650.
[180] D. Gong, B. Xu, Y. Zhang, Y. Guo, S. Yang, A similarity-based cooperative co-evolutionary algorithm for dynamic interval multiobjective optimization problems, IEEE Trans. Evol. Comput. 24 (1) (2019) 142–156.
[181] T.-C. Chiang, W.-H. Hsu, A knowledge-based evolutionary algorithm for the multiobjective vehicle routing problem with time windows, Comput. Oper. Res. 45 (2014) 25–37.
[182] Y. Xiang, Y. Zhou, Z. Zheng, M. Li, Configuring software product lines by combining many-objective optimization and SAT solvers, ACM Trans. Softw. Eng. Methodol. 26 (4) (2018) 1–46.
[183] G. Nicosia, S. Rinaudo, E. Sciacca, An evolutionary algorithm-based approach to robust analog circuit design using constrained multi-objective optimization, in: International Conference on Innovative Techniques and Applications of Artificial Intelligence, Springer, 2007, pp. 7–20.
[184] S. Kwag, S.-Y. Ok, Robust design of seismic isolation system using constrained multi-objective optimization technique, KSCE J. Civ. Eng. 17 (2013) 1051–1063.
[185] D. Zhan, H. Xing, Expected improvement for expensive optimization: a review, J. Global Optim. 78 (3) (2020) 507–544.
[186] J.-S. Pan, N. Liu, S.-C. Chu, T. Lai, An efficient surrogate-assisted hybrid optimization algorithm for expensive optimization problems, Inform. Sci. 561 (2021) 304–325.
[187] A.I. Forrester, N.W. Bressloff, A.J. Keane, Optimization using surrogate models and partially converged computational fluid dynamics simulations, Proc. R. Soc. A 462 (2071) (2006) 2177–2204.
[188] Q. Gu, Q. Wang, X. Li, X. Li, A surrogate-assisted multi-objective particle swarm optimization of expensive constrained combinatorial optimization problems, Knowl.-Based Syst. 223 (2021) 107049.
[189] Z. Song, H. Wang, B. Xue, M. Zhang, Y. Jin, Balancing objective optimization and constraint satisfaction in expensive constrained evolutionary multi-objective optimization, IEEE Trans. Evol. Comput. (2023).
[190] Z. Yang, H. Qiu, L. Gao, L. Chen, J. Liu, Surrogate-assisted MOEA/D for expensive constrained multi-objective optimization, Inform. Sci. 639 (2023) 119016.
[191] F.-F. Wei, W.-N. Chen, Q. Li, S.-W. Jeon, J. Zhang, Distributed and expensive evolutionary constrained optimization with on-demand evaluation, IEEE Trans. Evol. Comput. (2022).